

Chain of Thought Limits Lure Answers in Arithmetic

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Abstract

A variable named four can have value 6. We ask whether writing a chain of thought keeps a model from answering 4. We test matched arithmetic problems that differ in one name, across thirteen models and three sets of worked examples. Without a chain, eleven models show a naming effect on accuracy or lure answers. With a chain, 43 of 44 lure contrasts fall within the preregistered ± 2 -point margin. The exception is OLMo-2-1B-Instruct. On neutral problems, probes read the true value accurately at the step that writes it in 9 of 10 tested model-variable pairs; the exception has low probe accuracy at that step. Patching localizes the no-chain effect in Llama-3.2-3B to early layers at the variable’s defining equation. Reduced lure answers accompany accurate value readouts across the tested pairs. Yet the exception can state the correct value and still answer with the lure.

1 Introduction

In the problem

pen=1 + cup, cup=2 + 3; pen=?

pen has value 6. Renaming it four leaves the answer unchanged:

four=1 + cup, cup=2 + 3; four=?

The definition says 6; the name suggests 4. We call 4 the *lure value*. Because it is absent from the equations and differs from every variable’s value, answering 4 points to the name rather than to a copied digit.

Misleading identifiers and comments reduce accuracy on code tasks (Le et al., 2025; Lam et al., 2025; Le et al., 2026). Accuracy alone cannot locate the error: the model may misread the input, compute the wrong value, or answer incorrectly despite retaining the right value. Arithmetic with named variables gives us a known correct value to compare with the lure. Kudo et al.

(2026) use probes and patching to study when values become available during chain-of-thought (CoT) arithmetic. We add names that disagree with those values. This cue-conflict design, used in Stroop tasks and shape–texture studies (Stroop, 1935; Geirhos et al., 2019), gives the name and the computation distinct answers.

Writing a chain usually limits lure answers. Without CoT, names affect answers in eleven models. With CoT, almost every lure contrast meets our equivalence bound, although some accuracy effects remain. We then ask what distinguishes the exception. Across the probed model-variable pairs, low lure excess accompanies an accurate value readout at the step that writes it. This is an association; the readout need not drive the answer.

2 Study design

We use Kudo et al.’s single-digit arithmetic task: equations with addition or subtraction, followed by a query. Our main level requires two operations, with the queried variable defined before its dependency. Two models are also tested on all five levels (Appendix A). Each problem has matched versions differing in one variable name (Table 1). A congruent number word denotes that variable’s true value; an incongruent word denotes a lure. We vary the queried and intermediate variables separately. The lure cannot equal any variable’s value or input digit. Names occupy one token in every context, keeping twins’ token positions identical.

Models either write a chain that restates and evaluates each equation or answer directly. Every behavioral comparison uses three fixed sets of three worked examples. We evaluate thirteen checkpoints from the Llama, Gemma and OLMo families; the model list and tuning comparisons are in Appendices A and F. The main behavioral table follows the preregistered 90% mean neutral-CoT accuracy floor; the excluded model’s results

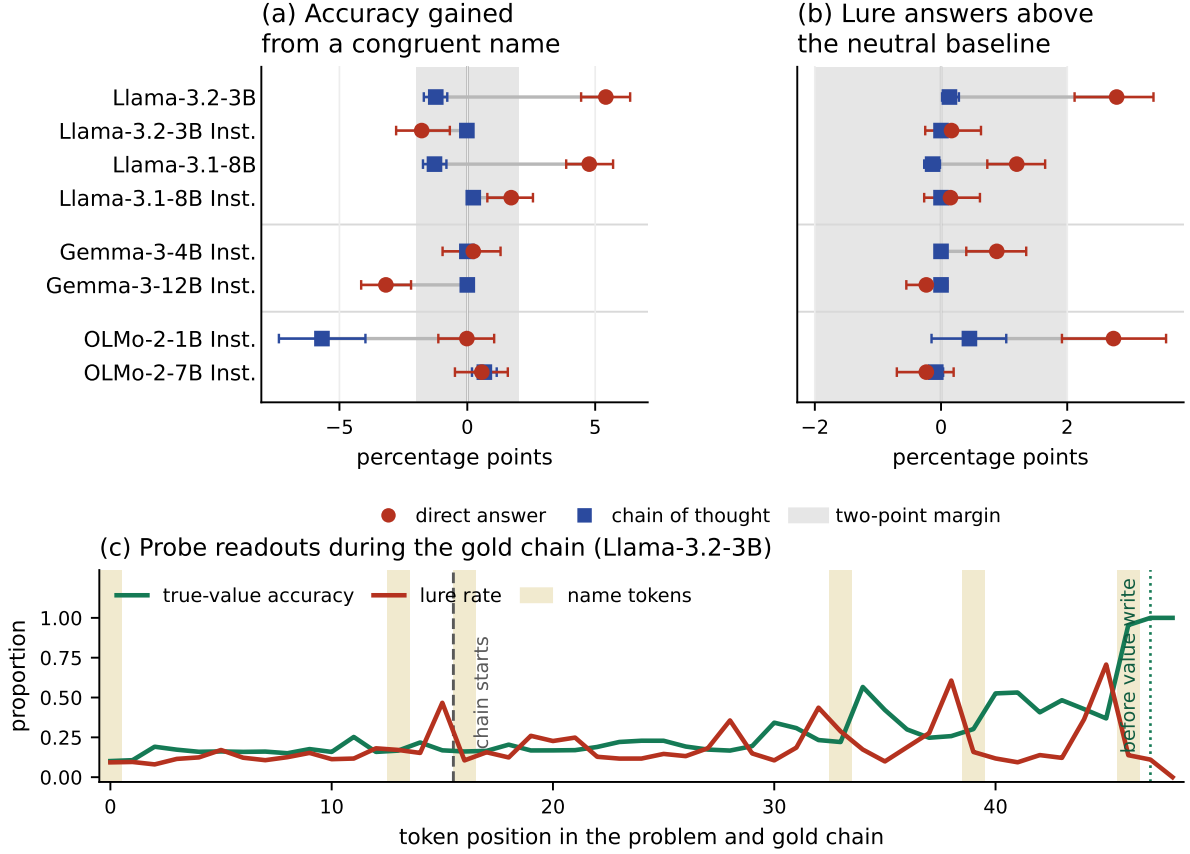


Figure 1: Eight panel models, queried-variable effects at level 3: (a) congruent-name accuracy gain and (b) lure excess, averaged over three demonstration sets with 95% intervals. Grey marks the two-point margin. (c) Maximum-over-layers true-value accuracy and lure rate at each token position, pooled over incongruent problems under a gold chain. The value readout concerns the queried variable. Shading marks name positions; the dotted line marks the position before its value.

Name	Problem (answer: 6)
Neutral	pen=1 + cup, cup=2 + 3; pen=?
Congruent	six=1 + cup, cup=2 + 3; six=?
Incongruent	four=1 + cup, cup=2 + 3; four=?

Table 1: Renaming the queried variable changes one cue while preserving the arithmetic. The incongruent name denotes 4, which appears nowhere else in the problem.

remain in the appendix.

Our main measure is *lure excess*: how often the incongruent problem receives the lure answer, minus how often its neutral twin receives that same digit. Subtracting this baseline matters when accuracy is low. We also report congruent-minus-neutral accuracy (*facilitation*) and neutral-minus-incongruent accuracy (*interference*). We call an effect reliable when its sign agrees across all three demonstration sets and its 95% cluster-bootstrap

interval excludes zero. We call an effect small enough to meet the equivalence bound when its 90% interval fits wholly inside ± 2 points. We resample matched problems with their seed replicates together (Appendix B).

A linear probe predicts a variable’s digit from a hidden state at one token and layer. Probes train on separate neutral problems, with three training seeds. A control assigns a random digit to each name, testing how well the same probe can learn name identity (Hewitt and Liang, 2019). Patching replaces activations at the name’s tokens with a matched twin’s (Zhang and Nanda, 2024). We report both a neutral-word control for disruption and an alternative-lure control for name-specific effects. Internal readouts use a force-decoded gold chain unless stated otherwise; they differ from the model’s freely generated answers.

Model	Regime	Neutral accuracy (%)	number word on the <i>queried</i> variable		number word on the <i>intermediate</i> variable	
			helps	lures	helps	lures
Gemma-3-12B-it	CoT	100 [100,100]	+0.0	+0.0	+0.0	+0.0
Gemma-3-12B-it	direct	78 [76,81]	-3.2*	-0.2	-3.7*	+0.4
Gemma-3-4B-it	CoT	100 [100,100]	-0.0	+0.0	-0.0	+0.0
Gemma-3-4B-it	direct	55 [48,59]	+0.2	+0.9*	+0.6	+0.3
Llama-3.1-8B-Instruct	CoT	100 [99,100]	+0.2	+0.0	+0.3	+0.0
Llama-3.1-8B-Instruct	direct	52 [45,65]	+1.7	+0.1	+5.2*	+0.2
Llama-3.1-8B	CoT	95 [86,100]	-1.3	-0.1	+3.8	-0.0
Llama-3.1-8B	direct	53 [41,62]	+4.8*	+1.2*	+3.8*	-0.0
Llama-3.2-3B-Instruct	CoT	100 [100,100]	-0.0	+0.0	-0.2	+0.0
Llama-3.2-3B-Instruct	direct	44 [40,50]	-1.8*	+0.2	+0.5	+0.5
Llama-3.2-3B	CoT	98 [97,100]	-1.2	+0.1	+0.9	-0.2*
Llama-3.2-3B	direct	22 [13,31]	+5.4*	+2.8*	+1.1	+1.8
OLMo-2-7B-Instruct	CoT	95 [93,97]	+0.7*	-0.1	-2.7	-0.0
OLMo-2-7B-Instruct	direct	54 [50,58]	+0.6	-0.2	+3.8*	-0.4

Table 2: Level 3, means over three demonstration sets (ranges in brackets). *Helps*: congruent-minus-neutral accuracy. *Lures*: lure excess. Both are percentage points. * marks a consistent sign across sets and a 95% interval excluding zero. CoT lure effects meet the two-point equivalence bound. The excluded low-accuracy model appears in Table 4; additional models and levels appear in Tables 7 and 9.

3 Results

3.1 Writing a chain limits lure answers

Table 2 shows the naming effects at the main level. Without CoT, a congruent name raises Llama-3.1-8B’s accuracy by +4.8 points, and a misleading name produces +1.2 points of lure excess on the queried variable. Across levels, the largest effects occur where direct-answer accuracy is near chance; the matched baseline prevents those arbitrary digit answers from being attributed to the name.

With CoT, 43 of 44 lure contrasts meet the equivalence bound. The exception is OLMo-2-1B-Instruct: its intermediate-variable lure excess is +3.2 points [+2.5, +3.9]. Its neutral CoT accuracy is 47%, compared with 77% under letter names. CoT therefore limits the name’s numeric influence without removing every naming effect: 15 accuracy contrasts remain reliable (Appendix B).

3.2 The value readout distinguishes the exception

We next compare readouts of the computed digit and the name’s digit. For Llama-3.2-3B, the intermediate value is partly decodable at the end of its

defining equation (neutral accuracy 0.49; selectivity 0.32). The lure can also be read near occurrences of the name. At the query, however, the name-identity control scores 0.97, so a high lure readout there is not evidence that the value has been bound to the variable. At the value-writing step and before the final answer, true-value probe accuracy reaches 0.98 or more and the lure readout falls to zero (Appendix E). Figure 1(c) shows the corresponding queried-variable readout.

Across the 10 probed model–variable pairs, we compare neutral accuracy at the layer with highest neutral selectivity. The 9 with true-value accuracy of at least 0.98 at the writing step have lure excess within 0.4 points of zero. Among pairs below the 0.9 threshold, the largest lure excess belongs to the exception model, with accuracy 0.22. This supports a link between a decodable value and reduced lure answers. Within individual problems, the probe–error association changes sign across fits. We therefore cannot use these readouts to explain each answer (Appendix E).

Merely stating the correct value does not explain the pattern. We split the model’s generated chains by whether they state that value and com-

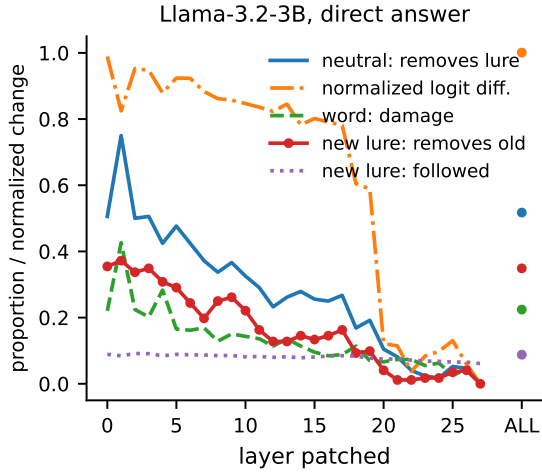


Figure 2: Llama-3.2-3B without CoT, queried variable: lure errors removed by the neutral-twin patch, alongside neutral-word damage and the alternative-lure control, by layer.

pare only twins falling in the same split. The exception still has +3.6 points of lure excess [+2.6, +4.7] across 2739 matched observations from 3 demonstration sets, despite stating the correct value. The full control is in Appendix D.

3.3 Patching localizes the no-chain effect

For Llama-3.2-3B without CoT, replacing the name’s activations with its neutral twin’s removes 52% of lure errors when all layers are patched (Figure 2). A span-by-layer sweep places the strongest effects at the defining equation in layers 0–7, where removal reaches 69% (Appendix K). The neutral-word control also damages 22% of correct answers. The patch identifies a region sensitive to the name, but the disruption limits what we can say about the computation it restores.

The Llama-3.1-8B controls do not distinguish a name-specific cause: neutral and alternative-lure patches remove errors similarly. Under a gold CoT, injecting the misleading name into a neutral run of the exception model moves 2.7% of answers to the lure, while the word control changes 0.4% at the same layer. These interventions test only the selected models and positions; they do not establish a common causal mechanism across models.

4 Related work

Kudo et al. (2026) locate intermediate-value readouts during arithmetic CoT; our competing names distinguish a computed value from a lexical cue. Earlier Stroop studies test a single next token

(Wang, 2026; Hu et al., 2026). Faithfulness studies perturb prompts, chains or parameters (Turpin et al., 2023; Lanham et al., 2023; Chen et al., 2025; Tutek et al., 2025); we combine a known conflicting cue with internal readouts. Positional copying and output priors can also explain arithmetic errors (Liu, 2026; Prabhakar et al., 2024); our copy control is reported in Appendix E. A companion code study (Anonymous, 2026) examines value writes in Python traces; this paper tests arithmetic lure answers and their association with value-step readouts.

5 Conclusion

Writing a chain limits lure answers in almost every tested arithmetic contrast. Across the probed pairs, this reduction accompanies an accurately decodable value at the writing step; stating the value in the chain alone is insufficient. In the exception, the chain itself can write the lure and pass it to the answer.

Limitations

The task uses short synthetic arithmetic chains and English number words. It does not establish how misleading identifiers affect real code. Identifier-style names were tested at one level only (Appendix D). Models are at most 12B parameters and receive raw completion prompts; instruction-tuned models’ chat templates and long-form reasoning modes are untested. Demonstration choice changes accuracy substantially, which is why every behavioral claim must hold across three sets.

Linear probes reveal decodable information, not necessarily the representation used to generate an answer. Most internal measurements supply the correct chain token by token. The small sample of models’ own chains is illustrative (Appendix E). The value-readout association holds across the probed pairs but is unresolved within instances and may reflect overall task competence. Patching changes mixed representations and can damage correct answers, limiting the causal conclusions. Finally, the chain repeats names and equations; the value-only format changes performance and cannot be treated as an equivalent intervention (Appendix H).

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Level	Problem (neutral names)	Steps	Held	Distr.
1	pen=1 + cup, cup=2; pen=?	1	1	0
2	pen=2 + 3, cup=1 + pen; cup=?	2	0	0
3	pen=1 + cup, cup=2 + 3; pen=?	2	1	0
4	pen=1 + cup, cup=2 + 3, box=4 + 5; pen=?	2	1	1
5	pen=1 + cup, cup=2 + box, box=1 + 2; pen=?	3	2	0

Table 3: The five levels of Kudo et al. (2026): additions or subtractions needed (*Steps*), variables unresolved when first read (*Held*), and unused equations (*Distr.*). At level 1 the intermediate value is stated rather than computed.

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A Task, models and replication

A problem contains single-digit variables, addition or subtraction, and a query. The five levels vary dependency order, depth and the presence of an unused equation (Table 3). Three same-level worked examples precede each test problem. The evaluated models are Gemma-3-12B-it, Gemma-3-4B, Gemma-3-4B-it, Llama-3.1-8B, Llama-3.1-8B-Instruct, Llama-3.2-3B, Llama-3.2-3B-Instruct, OLMo-2-1B-Instruct, OLMo-2-7B-Instruct, plus four derived checkpoints (Appendix F).

We reimplement Kudo et al. (2026)’s task and probe recipe rather than using their code. The shared choices are single-digit outputs, same-level demonstrations, a 10-way linear residual-stream classifier, full-batch SGD at 10^{-3} for 10,000 epochs, and their span-by-layer-window patching design. For replication we train on letter names and report t^* , t_{eq}^* , $Acc^{<CoT}$ and $Acc^{>CoT}$ at $\tau = 0.9$. The Llama-3.2-3B value-writing positions agree with their Table 3: equation 5 for the queried variable and 2 for the intermediate. Our additions are the conflicting names, matched

baselines, demonstration-set claim rule and equivalence bound.

On the letter condition, pre-chain probe maxima for Llama-3.2-3B are 30.4 and 47.9, compared with 17.8 and 33.2 reported by Kudo et al. (2026). The value-writing positions agree; the accuracy difference may reflect training details. Llama-3.1-8B’s intermediate value becomes decodable at equation 2, but its queried value becomes decodable at equation 3, earlier than Llama-3.2-3B’s equation 5. The larger model can therefore expose that value before writing its final evaluation. Full heatmaps appear in Appendix K.

B Equivalence checks

Following preregistration Amendment 3, equivalence requires a 90% cluster-bootstrap interval wholly inside the ± 2 -point margin (two one-sided 5% tests). We resample matched sets with their demonstration-seed replicates together, using 4,000 bootstrap draws. Other behavioral intervals remain at 95%. For CoT accuracy, 61 of 88 facilitation and interference contrasts meet the same equivalence bound. Of the 15 reliable accuracy contrasts, 5 are also equivalent within the margin and 10 are not; a detectable effect can still be small. The written-value control uses the same matched pairs for its excess estimate and interval; the exception model’s estimate pools all three demonstration sets and clusters on matched set.

C Tokenizer checks

We test candidate names in five tokenizer contexts and retain words that form one token without merging with a neighbor. For Llama 3, all ten number words and 95 of 112 neutral candidates pass. Gemma and OLMo use the same lists; runtime checks confirm equal token lengths and positions within matched groups. Spaces around binary operators prevent merges such as 1-two. No instance in the reported runs is excluded by the layout check.

D Additional naming and written-value controls

Written values. We split generated chains by whether they state the target’s true value. An observation enters the excess estimate only when its neutral twin falls in the same split. The estimate and its 95% interval use those same paired observations. Pooled over eligible cells, lure excess is

Model	Regime	Neutral accuracy (%)	number word on the <i>queried</i> variable		number word on the <i>intermediate</i> variable	
			helps	lures	helps	lures
OLMo-2-1B-Instruct	CoT	47 [44,51]	-5.7*	+0.4	+7.0*	+3.2*
OLMo-2-1B-Instruct	direct	34 [30,38]	-0.0	+2.7*	+3.0*	-0.2

Table 4: The level-3 model below the preregistered 90% mean neutral-CoT accuracy floor, excluded from the main behavioral table. Columns and claim markers follow Table 2.

Contrast	Cells	Equivalent	Reliable
CoT lure	44	43	3
CoT accuracy	88	61	15
Direct lure	44	31	13

Table 5: Equivalence within the two-point margin uses 90% intervals; reliability uses 95% intervals excluding zero and the same sign across all three demonstration sets. Each accuracy cell is one facilitation or interference contrast.

+0.08 points in 132 cells where the value is stated and -0.10 in 27 where it is not. The exception persists in the stated-value split (main text), so this split does not explain its effect.

Identifier-style names. At level 3, four models receive names such as q4 instead of four, under three demonstration sets. Without CoT, queried-variable lure excess ranges from -0.2 to +0.2 points; 0 of 4 cells are reliable. For number words on the same models, the range is -0.2 to +2.8, with 3 of 4 reliable. The numeric lure effect is therefore stronger for number words in this comparison; congruent identifier names can still improve accuracy.

Word-class effects. With CoT at level 4, a number word on the queried variable costs Llama-3.2-3B 9.9 accuracy points whether the name agrees with the value or not. Those errors restate the wrong equation. This is distinct from answering the digit the name denotes, which is why the lure result cannot be read as removal of every naming effect.

Unused variables. A number word on the level 4 distractor changes accuracy by at most 1.2 points. Its no-CoT lure excess ranges from +2.1 to +2.3; 2 of 2 cells are reliable. With CoT, the range is +0.0 to +1.0, with 0 of 2 reliable. At the answer position, probe lure mass is 0.004 for a computed variable and 0.118 for the distractor; the latter’s true value has probe accuracy 0.04.

E Full sweeps and selectivity

Mixed-effects check. We compare the paired bootstrap with a mixed logit model, $\text{answered_lure} \sim \text{incongruent} + \text{C}(\text{seed})$, with a random intercept for matched set. Each twin’s outcome refers to the same lure digit. Both procedures use the same fixed subsample of 600 sets per cell. Of 57 fitted cells, 50 agree on whether the interval excludes zero; 11 additional cells have no outcome variation. The mixed fit supports an effect in 5 cells where the bootstrap does not and withdraws support in 2 cells where it does (Gemma-3-12B-it, value-only chain, queried variable; Llama-3.1-Nemotron-Nano-8B, CoT, intermediate variable). Subsampling also removes support from 1 starred Table 2 cell under both procedures (Gemma-3-4B-it, no CoT, queried variable). The mixed model uses variational Bayes, so its interval is an approximate posterior interval. We retain the full-data bootstrap as the primary estimate on the reported percentage-point scale.

Instance-level association. Within each model, level, regime, variable and position, we regress lure errors on the probe margin, $\log p_{\text{true}} - \log p_{\text{lure}}$, with standard errors clustered on matched set. The layer is selected by neutral accuracy. Of 57 fits, 27 have $p < .05$: 22 coefficients are negative and 5 positive. The direction is not consistent, and 4 positive results come from the model supplying most fits. This leaves the instance-level association unresolved, even though the aggregate pattern across model–variable pairs is clear.

Lure index. For comparison with shape–texture studies (Geirhos et al., 2019), we also compute lure answers divided by lure plus true answers, excluding congruent trials. Across 18 level 3 model–variable pairs, the index ranges from 0.02 to 0.38 without CoT and from 0.00 to 0.16 with CoT. The main text uses paired excess because the index

Target	Read	Errors	Neutral removed	Word dmg.	Alt. removed	Alt. followed
v1	value	0	—	1.6	—	0.0
v1	answer	0	—	1.6	—	0.0
v2	value	28	92.9	7.3	82.1	4.0
v2	answer	3	100.0	0.8	100.0	0.6

Table 6: OLMo-2-1B-Instruct, CoT, 500 matched pairs per target, patching name tokens at all layers. Removed: percentage of baseline lure predictions removed by the neutral-twin patch. Word damage: percentage of correct neutral predictions damaged by the word control. Alt. removed: baseline lure predictions removed by the alternative-lure patch. Alt. followed: percentage following that source’s lure. Rates are percentages; — means undefined. These readouts force the gold chain, so their baseline errors differ from errors in the model’s own generated chains.

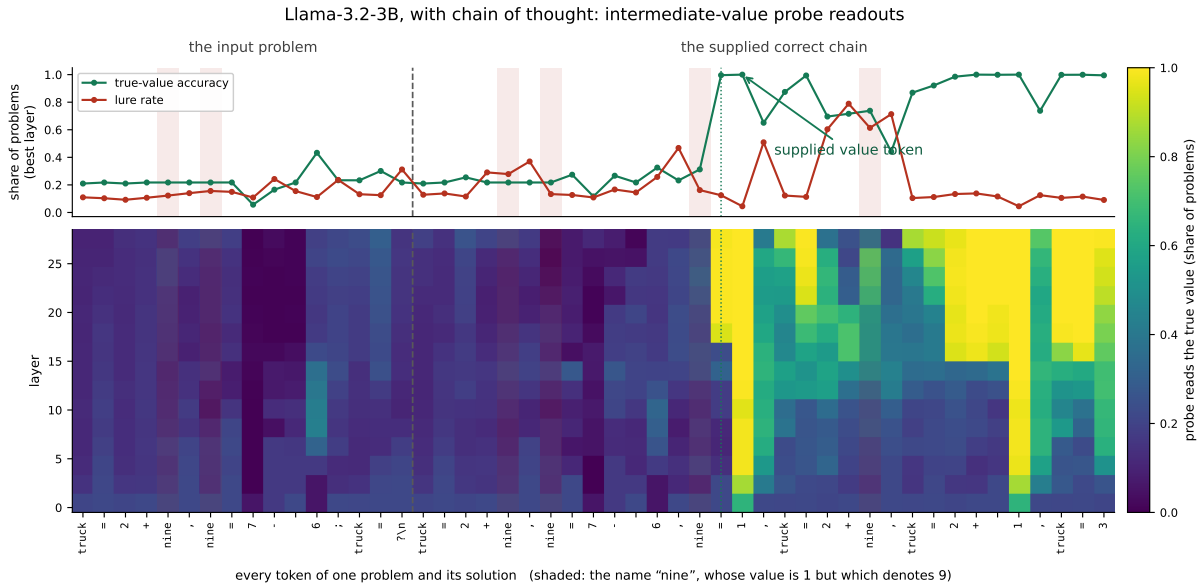


Figure 3: In the layout of Kudo et al.’s Figure 2. Per-token probes for the incongruent condition, Llama-3.2-3B, level 3, CoT. Top: how often the best-layer probe reads the true value (green) or the lure (red). Bottom: true-value accuracy by layer. The displayed tokens are from one example; the rates pool problems with different values and names. Dashed line: start of the supplied correct chain; dotted line: position before the target value; arrow: the supplied value token. Shading marks the name’s tokens.

does not subtract how often the neutral model answers that digit.

Positional-copy control. Across 4110 generated CoT lure errors over models, levels and demonstration sets, the number immediately preceding the answer is the lure in 13% of cases and the target’s true value in 11%. A trailing lure is therefore absent from many errors. This check constrains the positional-copy explanation of Liu (2026), but does not identify what was copied in each case.

Probes report true-value accuracy, lure rate, lure mass and margin, with name-identity control accuracy and selectivity (Hewitt and Liang, 2019). P1 is the name token, P2 the end of its defining

equation, P3 the query, P4 the position before its value is written, and P5 the position before the final answer. P1 is a reference, not evidence for a computed value. At P3 the control is near 1, so a lure readout there may reflect name identity. Figure 3 shows the CoT token sweep; Appendix K contains the other conditions and direct regime. Letter-trained replication uses the metrics in appendix A.

F How the models were tuned

We compare six checkpoints on the Llama-3.1-8B architecture: pretrained, instruction-tuned, single-task LoRA, multi-task LoRA, TIES-merged adapters and reasoning-tuned. Adapters are folded

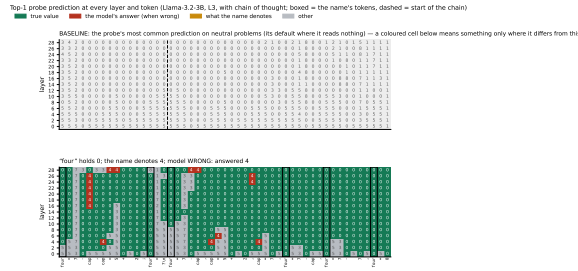


Figure 4: In the layout of Kudo et al.’s Figure 3, for problems this model answers with the lure. Each cell is the probe’s top-1 prediction at that layer and token: green the variable’s true value, red the value the model answered, amber the value its name denotes, grey anything else. The first panel is the probe’s most common prediction on neutral problems, which is what it emits where it reads nothing; cells carry information only where they differ from it.

into the weights before evaluation. Shared tokenizers and identical matched problems permit paired comparisons (Table 7). A pretrained/instruct Gemma pair supplies a second family comparison.

Tuning	Model	Queried	Intermediate	Acc.
base	Llama-3.1-8B	+1.20*	-0.02	53.4
instruct	+ instruct	+0.15	+0.22	52.2
single-task finetune	+ task LoRA	+0.05	+0.23	52.2
multi-task finetune	+ multi-task LoRA	-0.58*	-0.08	46.1
weight merge	+ TIES merge	-0.10	+0.10	51.9
reasoning-tuned	Nemotron-Nano-8B	+1.35*	+0.33	39.4
base	Gemma-3-4B	+0.08	+0.02	60.2
instruct	+ instruct	+0.88*	+0.33	54.9

Table 7: Level 3, no CoT, lure answers above the matched twin’s rate, mean over three demonstration sets. Acc. is neutral accuracy. * as in Table 2. The bottom block repeats the pretrained-to-instruct step in a second family, Gemma-3-4B.

The pretrained and reasoning-tuned Llama checkpoints have reliable positive queried-variable lure excess without CoT (+1.20 and +1.35 points). The three task-tuned variants differ from the instruction-tuned checkpoint by at most 0.73 points across 6 comparisons. These are observed differences, not tests of checkpoint-to-checkpoint equivalence. Every checkpoint’s CoT lure excess is at most 0.20 points in magnitude.

Task accuracy complicates interpretation. The multi-task model’s reliable negative excess (-0.58) accompanies lower neutral accuracy (46.1%, versus 52.2%). The reasoning-tuned checkpoint has the lowest accuracy (39.4%). It also requires a system instruction to enable long-form reasoning; our raw-completion prompts leave that mode off. The result concerns the checkpoint under this prompt,

not reasoning mode in deployment.

Only the instruct Gemma effect passes the claim rule (+0.88, versus +0.08 pretrained). The difference is untested and both estimates lie within the margin, so the pair cannot establish where in training name interference arises.

G Probe recipe and patching scope

We check two implementation choices inherited from Kudo et al. (2026): the probe optimizer and the name occurrences included in a patch.

Recipe. We refit the level-3 probes for Llama-3.2-3B with L-BFGS, and with SGD on standardized inputs, against the reported recipe.

Position	Variable	SGD (reported)	L-BFGS	standardized
defining equation	queried	0.184	0.190	0.183
defining equation	intermediate	0.496	0.422	0.420
query	queried	0.183	0.316	0.280
query	intermediate	0.271	0.366	0.393
value step	queried	1.000	1.000	1.000
value step	intermediate	0.982	0.962	0.972
answer	queried	1.000	1.000	1.000

Table 8: Best-layer accuracy for the true value, neutral instances, under three probe recipes.

All three recipes give the same qualitative result: the value is decodable where the chain writes it (0.96–1.00) and not at the end of its defining equation (0.18–0.50). The query position moves, from 0.271 under the reported recipe to 0.393 under standardization, within a range of 0.18–0.39 over recipes and variables. We already report that position as the presence of the name rather than a decoded value, because its control task scores 0.97, so this variation does not change the interpretation of that position.

Scope. Patching every occurrence of the name includes the copy the model writes in its own output. In the direct regime, restricting the patch to the name’s occurrences in the prompt leaves the intermediate variable essentially unchanged, since its name never appears in the answer: removal is 43% at layers 0–3 under both scopes. For the queried variable the best windows remove similar shares (52% at layers 4–7 prompt-only, against 53% at layers 0–3), the prompt-only patch removes less in the later windows, and the two disagree most in the first window, where the prompt-only patch removes 3% and damages 76% of correct answers. That is mechanical: the queried variable’s name also sits at the position we read, so patching its prompt copies alone leaves the

Top-1 probe prediction over the model's own chain (Llama-3.2-3B, L3, with chain of thought; probes trained at the value step of "v1" on neutral problems; 8 instances)

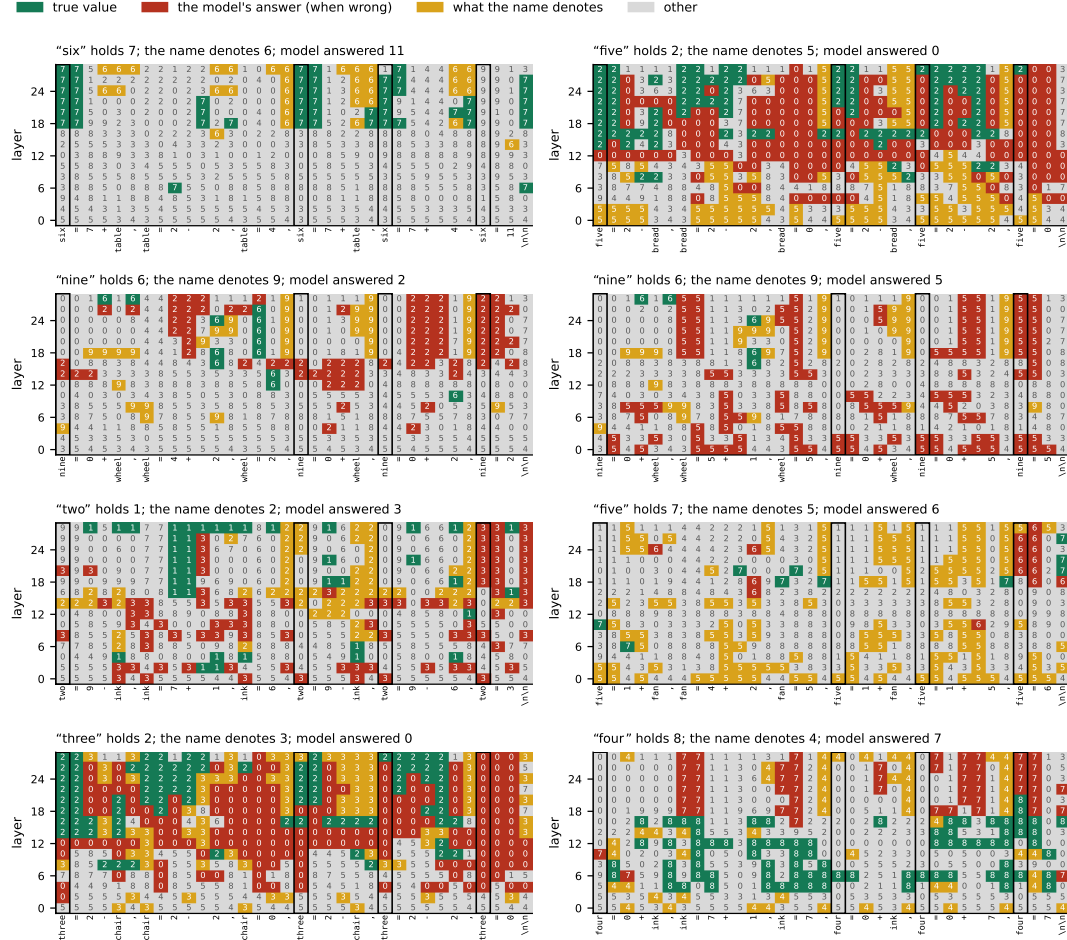


Figure 5: Probes trained at the value step, applied to the model’s own generated chain rather than to a force-decoded gold chain, for the 8 level-3 incongruent problems that Llama-3.2-3B answered wrongly under CoT (the first 8 of 108 wrong answers in 2,000 instances). Probes are trained on neutral problems at the step that writes the variable’s value and applied at every generated token and every second layer; colours as in Figure 4, boxed columns the name’s own tokens. This small sample contains no lure answers and is illustrative rather than an estimate of their rate.

model reading a name whose early representation no longer matches. The localization we report does not depend on the choice.

H A chain that writes values without equations

Our chain restates each equation before substituting into it, as Kudo et al.’s does. Their Tables 5–6 also report a format that writes only the sub-results, $\text{cup}=5$, $\text{pen}=6$. That format is the arithmetic counterpart of a code trace, which writes a variable’s value and never the expression that produced it, so we ran it on the panel at level 3 with three demonstration sets, with two predictions registered in advance for both Llama models: that the written-lure excess at the intermediate’s value

step would exceed the 2-point margin, and that the value would still be decodable there (probe accuracy at least 0.85), making any failure one of read-out.

Both predictions fail as registered. Llama-3.1-8B writes the lure at the intermediate’s value step +0.15 points above its twin, inside the margin. Llama-3.2-3B is at +9.20, which passes our claim rule, but across the three demonstration sets that is +1.7 / +24.2 / +1.7, so one set carries most of it. More informative is what the probes say. At the step that writes the value, the true value is decodable at 0.81 and 0.59 for Llama-3.1-8B and at 0.61 and 0.79 for Llama-3.2-3B, against 1.00 and 0.99 for the same model under the full chain. Neutral accuracy falls with it, to 54% and 31%.

Removing the equations lowers both accuracy and true-value probe readouts. It therefore fails to isolate a readout error while preserving those measures. This result does not support a shared explanation for value-only arithmetic chains and code traces.

I Additional levels, models and demonstration sets

Table 9 repeats Table 2 for the other four levels. The pretrained Gemma-3-4B, the base half of the Gemma pair, is reported in Table 7 rather than in Table 2.

Data, compute and software. Each level has 2,000 matched sets, each rendered in every condition, and 10,000 separately generated problems for training probes; every SGD probe is trained with 3 seeds. Every run here is inference and the probes are linear; the three adapter checkpoints of Appendix F were trained in a separate project, and that compute is not counted below. The recorded allocation time totals at most 178 GPU-hours on NVIDIA A30, H100 and H200 cards, including idle time and shared workers that also ran companion-code experiments. The models are from the Llama 3 (Grattafiori et al., 2024), Gemma 3 (Gemma Team, 2025) and OLMo 2 (Team OLMo, 2024) families, with the reasoning-tuned checkpoint from Bercovich et al. (2025), each used under its release licence for research. Models run in PyTorch (Paszke et al., 2019) with Transformers (Wolf et al., 2020), and the probes are trained in PyTorch; the L-BFGS refit of Appendix G uses scikit-learn (Pedregosa et al., 2011), and the mixed-effects check uses statsmodels (Seabold and Perktold, 2010).

Per-set values, the pooled estimate and the claim rule are released with the code, together with the level-4 irrelevant-lure control, the lure-distance analysis, the teacher-forced lure rates, and the error taxonomy behind the word-class effect.

J Additional value readouts

We extend the value-step comparison to Llama-3.2-3B-Instruct, Llama-3.1-8B-Instruct and Gemma-3-12B-Instruct. Each probe trains on 10,000 separate neutral problems with three probe seeds, using demonstration set 7. Table 10 reports the added pairs; the main comparison includes them. We select layers by neutral selectivity, as

Model	Level	Regime	Neutral accuracy (%)	number word on the queried variable		number word on the intermediate variable	
				helps	lures	helps	lures
Llama-3.1-8B	1	CoT	99 [98,100]	+0.0	+0.0	+0.3	+0.0
Llama-3.1-8B	1	direct	74 [38,99]	+16.1*	+11.2*	+8.8*	+1.3
Llama-3.2-3B	1	CoT	97 [95,100]	+1.1	+0.1	-2.4	+0.3
Llama-3.2-3B	1	direct	46 [21,93]	+0.6	+5.9*	-3.9*	+2.6*
Llama-3.1-8B	2	CoT	100 [100,100]	-0.1	+0.0	-0.0	+0.0
Llama-3.1-8B	2	direct	36 [33,37]	+11.8*	+0.7	+2.6	+2.2
Llama-3.2-3B	2	CoT	94 [90,98]	+3.7	+0.5*	+1.0*	-0.1
Llama-3.2-3B	2	direct	29 [27,30]	+10.8*	+2.5*	+2.6*	+2.2
Llama-3.1-8B	4	CoT	88 [63,100]	+1.0	-0.0	-2.3	+0.2
Llama-3.1-8B	4	direct	28 [26,31]	+13.7*	+5.8	+6.0*	+0.2
Llama-3.2-3B	4	CoT	64 [26,85]	-10.2*	+0.7	-2.6*	+0.4
Llama-3.2-3B	4	direct	20 [16,23]	-1.6	+1.9	-0.2	+1.2*
Llama-3.1-8B	5	CoT	65 [43,100]	+1.2	-0.3	+0.1	-0.1
Llama-3.1-8B	5	direct	20 [13,26]	+20.2*	+20.5*	+0.9	+0.4
Llama-3.2-3B	5	CoT	69 [48,97]	-1.4	+0.1	-2.5	-0.1
Llama-3.2-3B	5	direct	12 [11,12]	+4.7*	+3.5*	+1.1*	+0.5

Table 9: As Table 2, for levels 1, 2, 4 and 5 (Llama-3.2-3B and Llama-3.1-8B). Level 1 is the no-computation control: the intermediate variable’s value is stated rather than computed.

in the original analysis. The misleading-name accuracy uses the same selected layer. This extension tests the aggregate association; it does not resolve the within-problem link.

We also probe every token for Gemma-3-4B-Instruct (Figures 6 and 7), using 4,000 independent neutral training problems and 1,000 test problems per condition. The tokens shown come from one example; the curves and heatmaps pool problems with different names and values. Maximum-over-layers curves describe where a digit can be read; they do not identify the representation used to answer. On neutral problems, the best pre-chain accuracy is 20.0% for the queried digit and 20.0% for the intermediate digit. The queried readout first exceeds 90% at token 14; the intermediate readout first exceeds 90% at token 14. Token zero is the first token of the chain; negative positions are in the prompt.

Model	Variable	Layer	Neutral	Control	Misleading
Llama-3.2-3B-Instruct	queried	20	99.0	80.0	98.0
Llama-3.2-3B-Instruct	intermediate	20	99.0	80.0	98.0
Llama-3.1-8B-Instruct	queried	20	99.0	80.0	98.0
Llama-3.1-8B-Instruct	intermediate	20	99.0	80.0	98.0
Gemma-3-12B-it	queried	20	99.0	80.0	98.0
Gemma-3-12B-it	intermediate	20	99.0	80.0	98.0

Table 10: Additional probes at each variable’s value-writing position. Neutral and misleading columns are true-digit accuracy; control is neutral name-identity accuracy. Rates are percentages, averaged over three probe seeds. The layer maximizes neutral accuracy minus control accuracy.

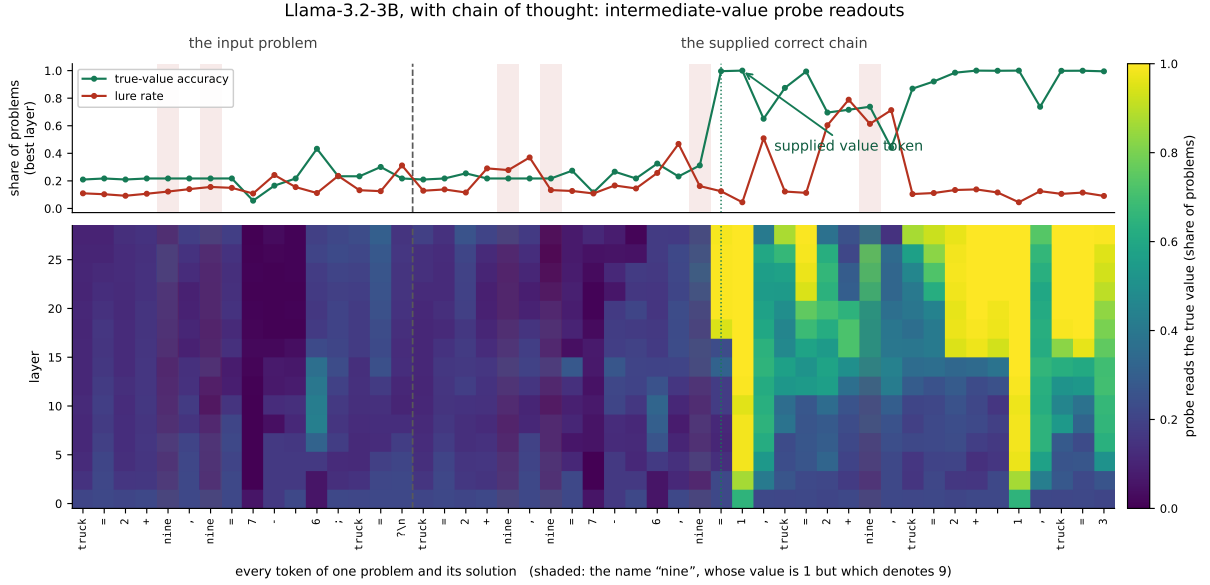


Figure 6: Gemma-3-4B-Instruct, queried variable, misleading-name problems under the correct chain. Top: maximum-over-layers true-value accuracy and lure rate; bottom: true-value accuracy by layer. Shading marks the variable’s name; the dotted line marks the position before its value; the arrow marks the supplied digit itself.

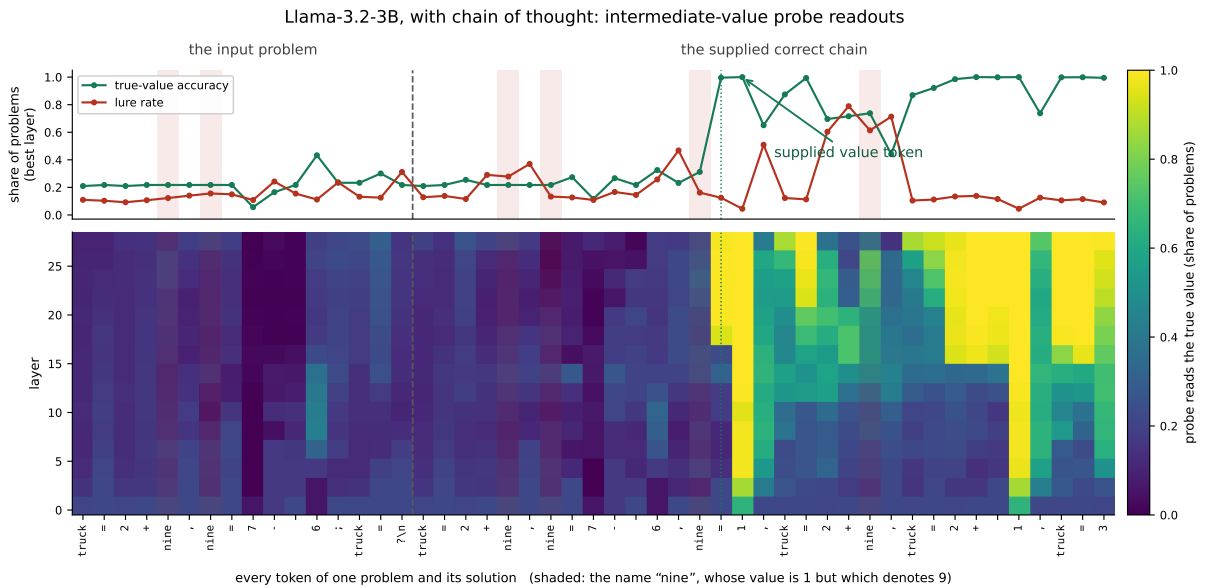


Figure 7: The same Gemma token sweep for the intermediate variable. Both figures use neutral-trained probes; no misleading-name problems enter training.

K Every figure at full size

Each figure below is reproduced as a full-width figure on a portrait page, within the ACL margins.

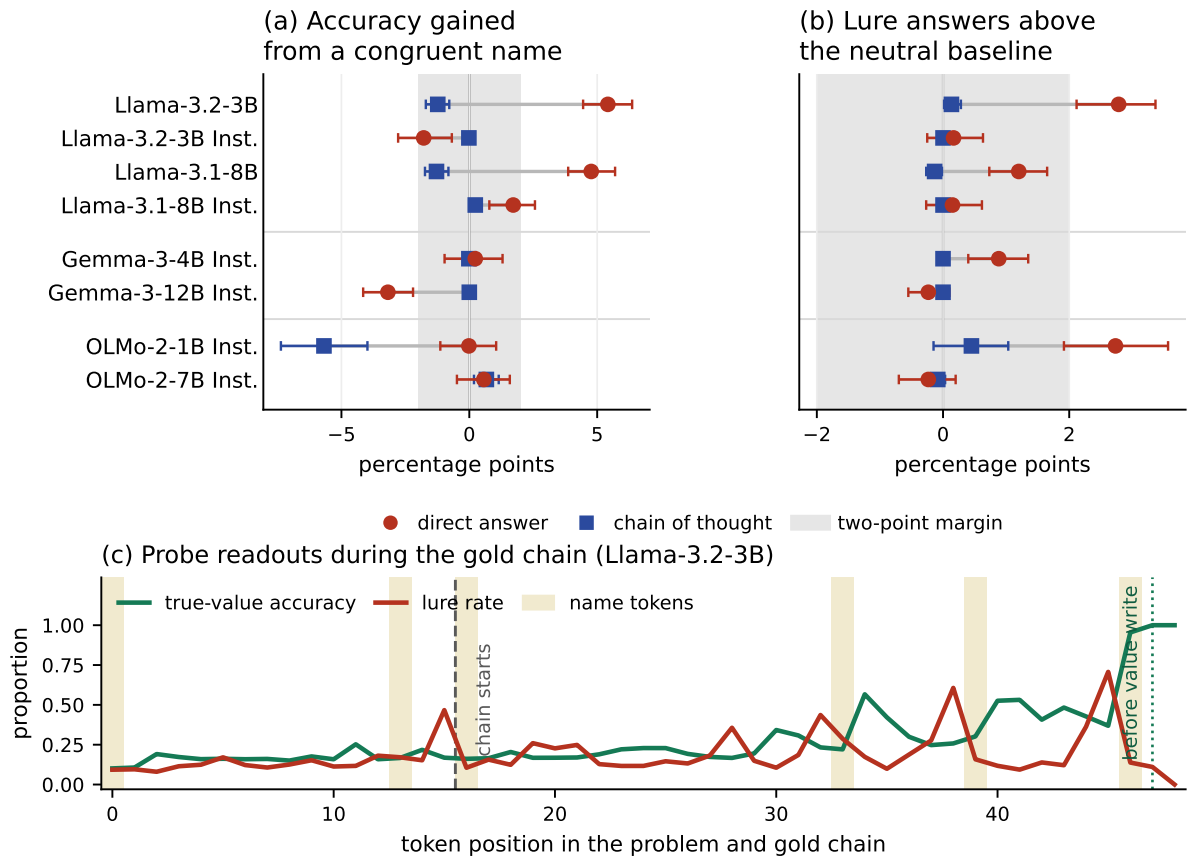


Figure 8: Figure 1 at full size.

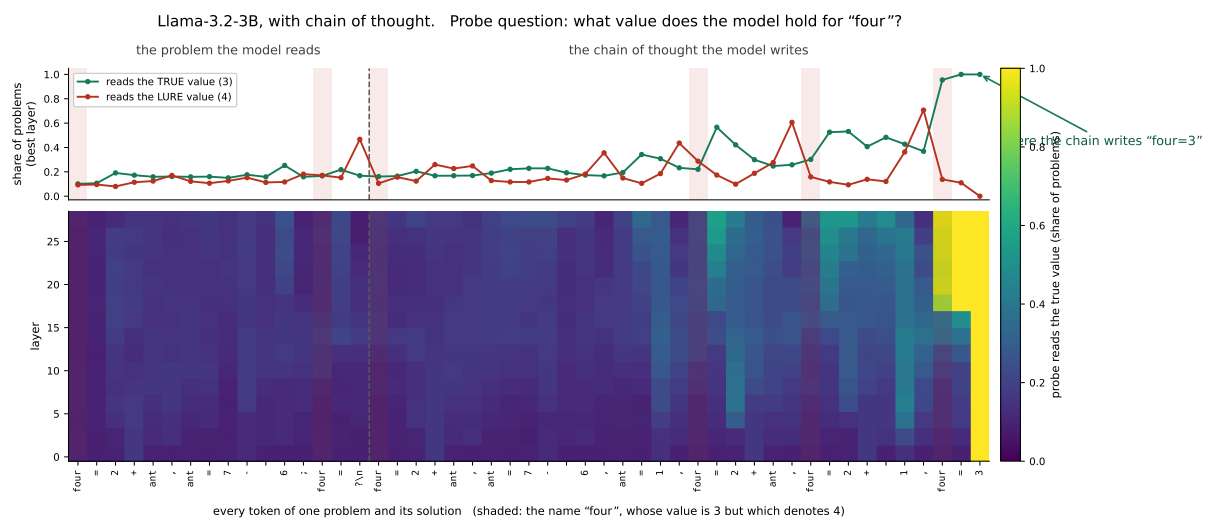


Figure 10: Per-token probes, chain-of-thought regime, queried variable.

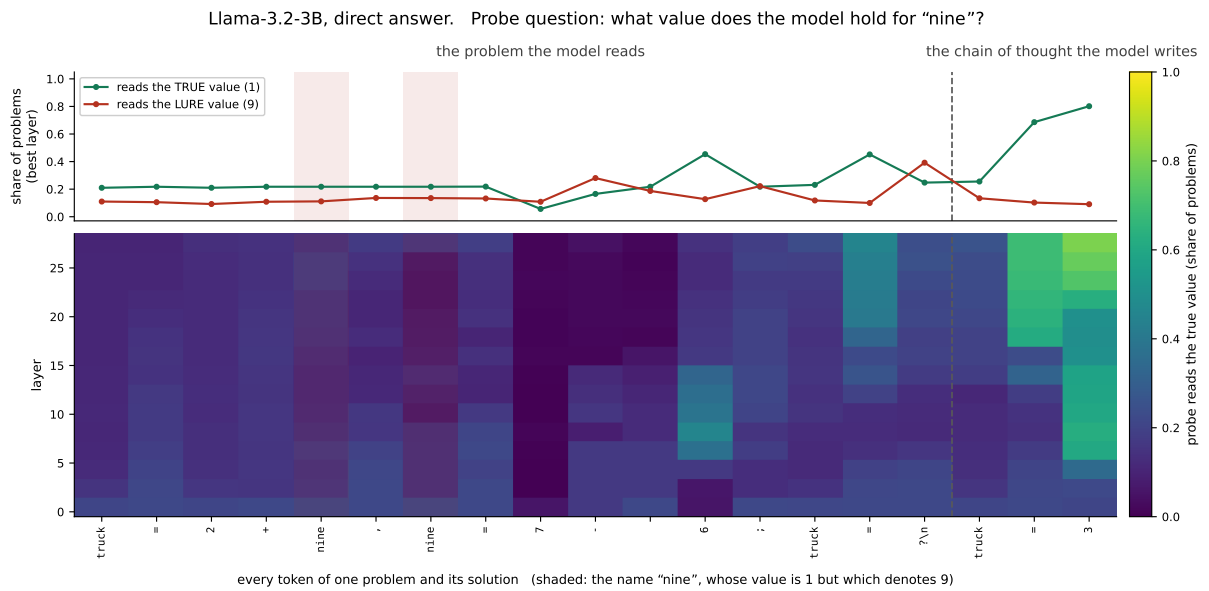
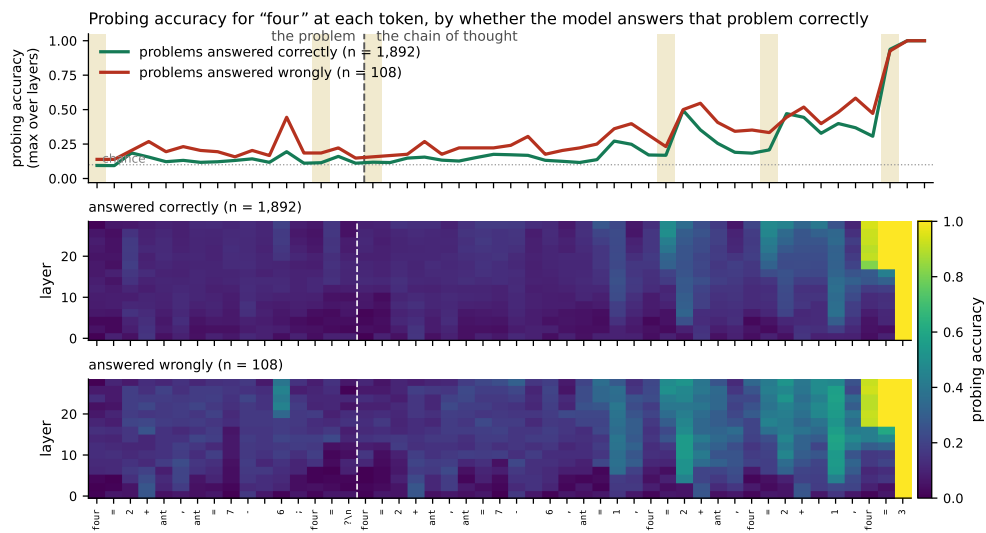


Figure 11: Per-token probes, direct regime, intermediate variable.



Probes read the variable's true value; states come from a forced pass over the correct chain, so the lower panel asks what the state holds for problems this model gets wrong unaided.

Figure 12: Probing accuracy split by whether the model answers the problem correctly on its own, with the layer sweep for each half.

Top-1 probe prediction at every layer and token (Llama-3.2-3B, L3, with chain of thought; boxed = the name's tokens, dashed = start of the chain)

■ true value
 ■ the model's answer (when wrong)
 ■ what the name denotes
 ■ other

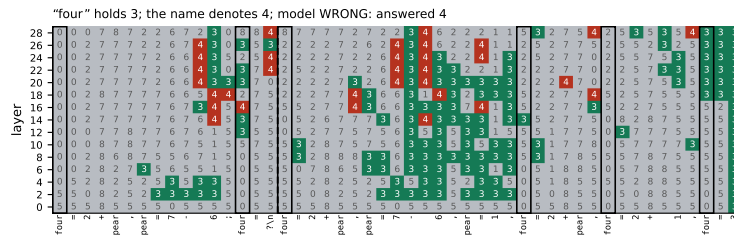
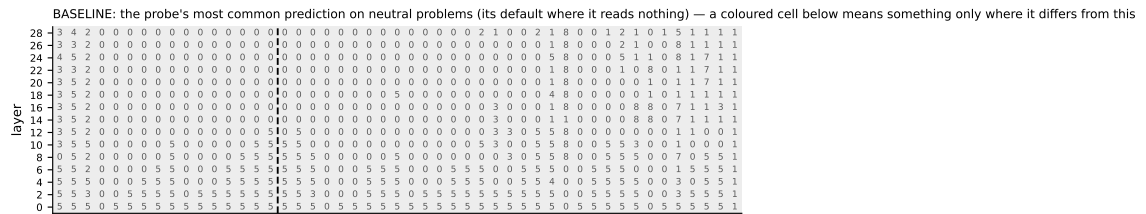


Figure 14: A second problem answered with the lure.

Top-1 probe prediction at every layer and token (Llama-3.2-3B, L3, with chain of thought; boxed = the name's tokens, dashed = start of the chain)

■ true value
 ■ the model's answer (when wrong)
 ■ what the name denotes
 ■ other

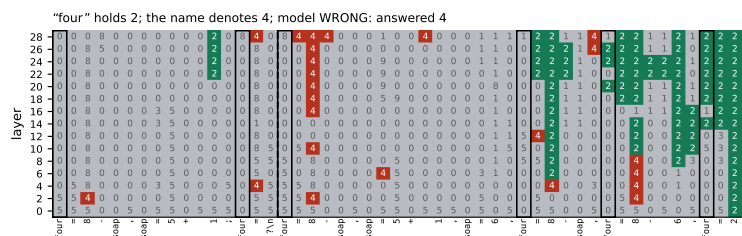
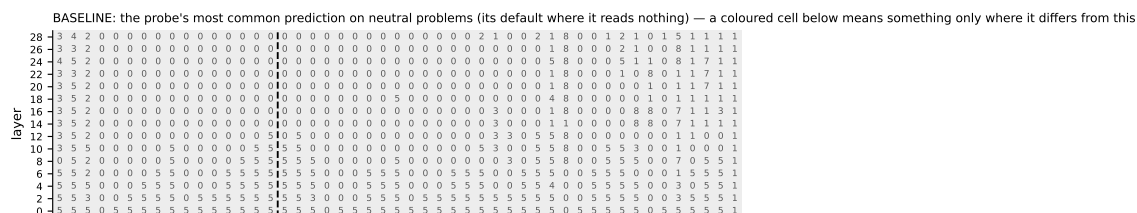


Figure 15: A third problem answered with the lure.

Top-1 probe prediction at every layer and token (Llama-3.2-3B, L3, with chain of thought; boxed = the name's tokens, dashed = start of the chain)

■ true value
 ■ the model's answer (when wrong)
 ■ what the name denotes
 ■ other

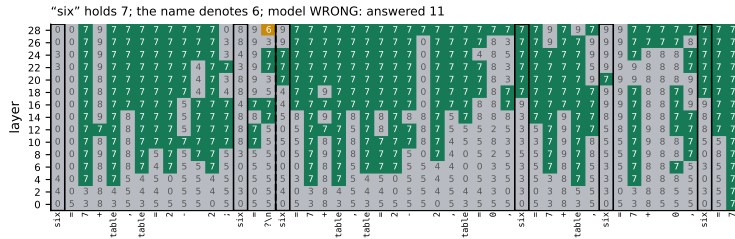
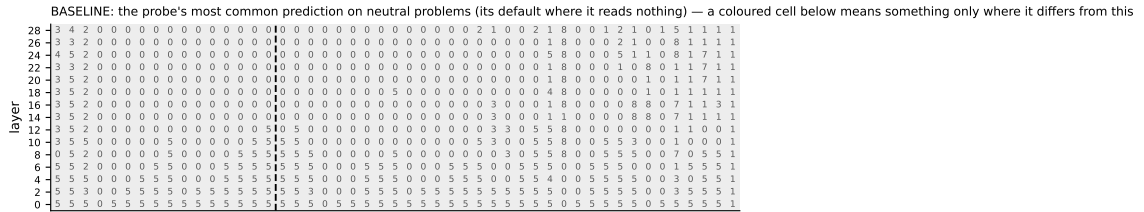


Figure 16: As above, on a problem answered wrongly for another reason.

Top-1 probe prediction at every layer and token (Llama-3.2-3B, L3, with chain of thought; boxed = the name's tokens, dashed = start of the chain)

■ true value
 ■ the model's answer (when wrong)
 ■ what the name denotes
 ■ other

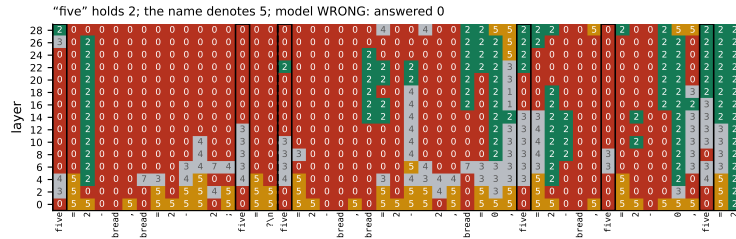
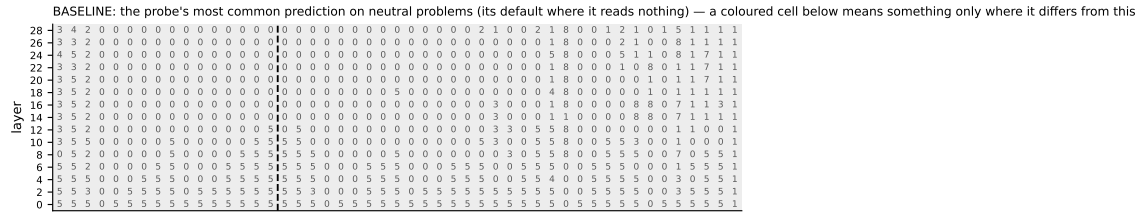
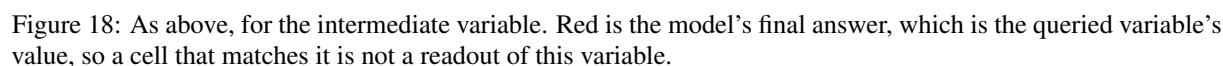
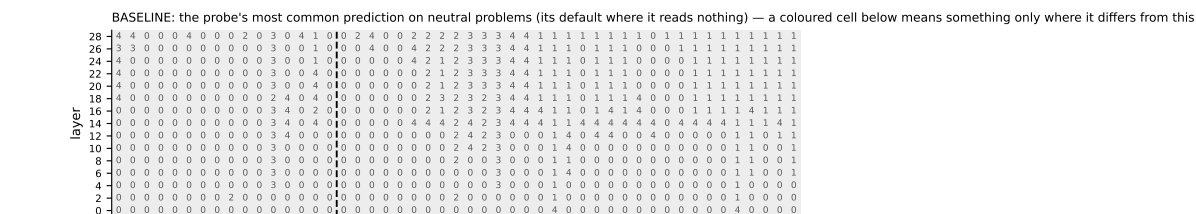


Figure 17: A second such problem.



Top-1 probe prediction at every layer and token (Llama-3.2-3B, L3, with chain of thought; boxed = the name's tokens, dashed = start of the chain)

■ true value
 ■ the model's answer (when wrong)
 ■ what the name denotes
 ■ other

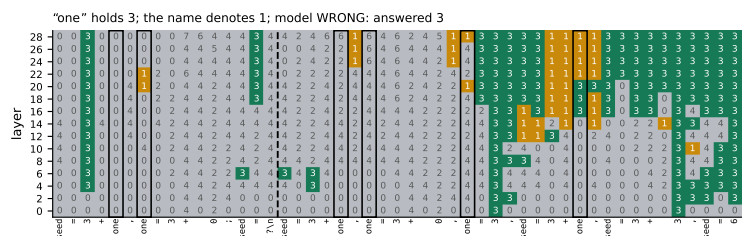
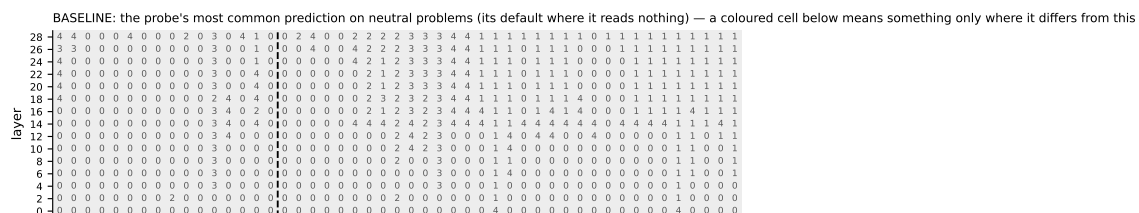


Figure 19: A second problem, intermediate variable.

Llama-3.2-3B, level 3, with chain of thought: per-token probes for the value of the intermediate variable (trained on neutral problems)

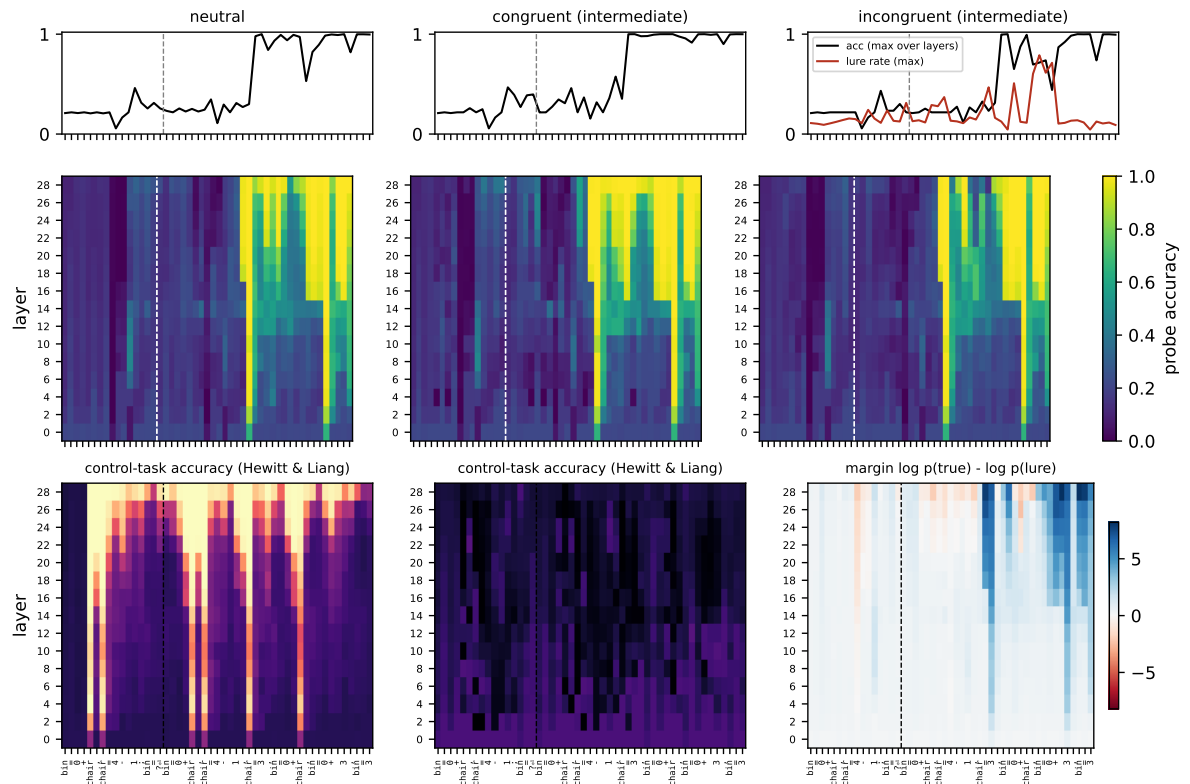


Figure 20: Accuracy heatmaps by token and layer for the neutral, congruent and incongruent conditions, in the layout of Kudo et al.’s Figure 2.

Llama-3.2-3B, level 3, direct answer: per-token probes for the value of the intermediate variable (trained on neutral problems)

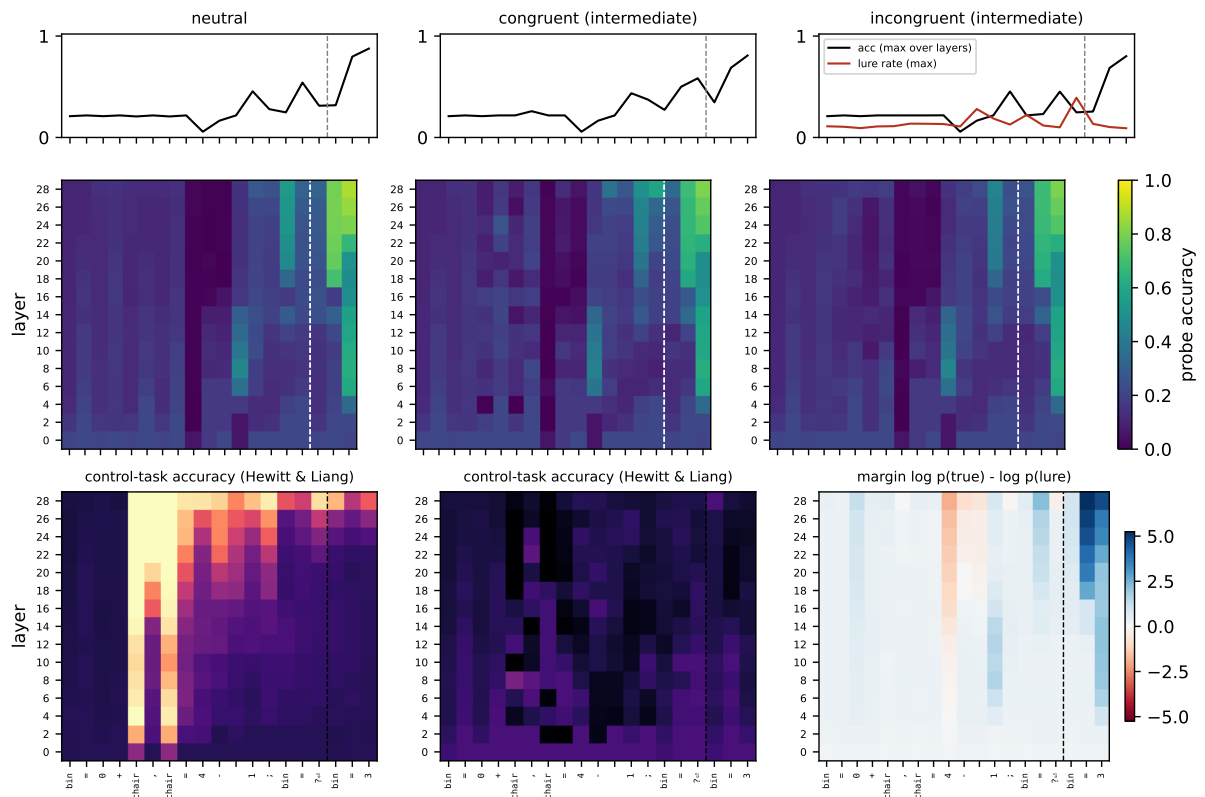


Figure 21: The same heatmaps in the direct regime.

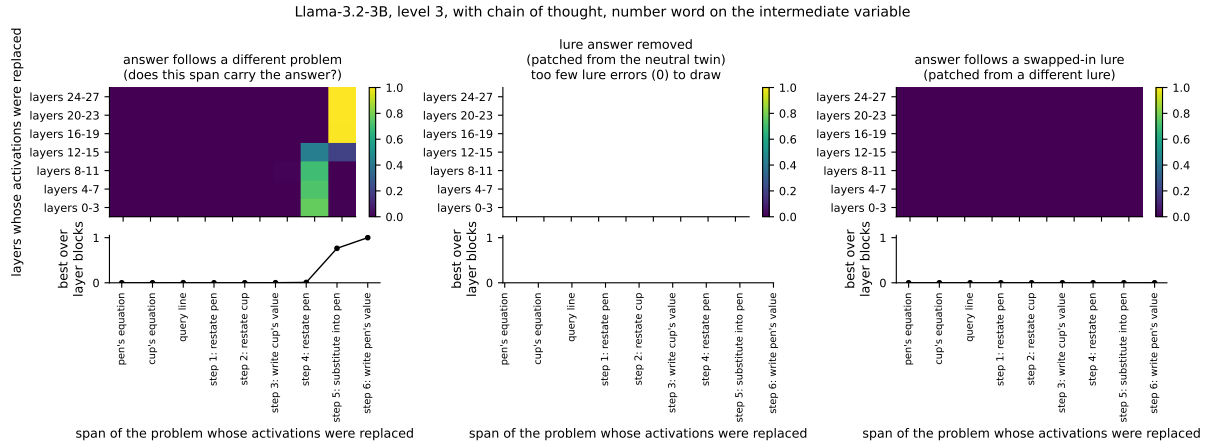


Figure 22: Span-by-layer-window patching grids, three sources (the lure-removal panel is empty: with CoT there are too few lure errors to draw), after Kudo et al.’s Figures 5–6.

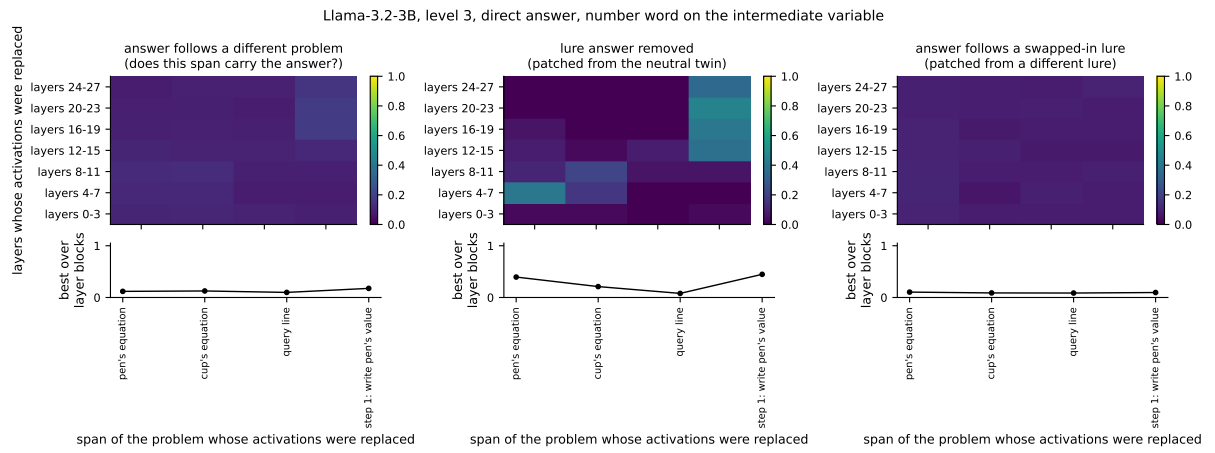


Figure 23: The same grids in the direct regime.

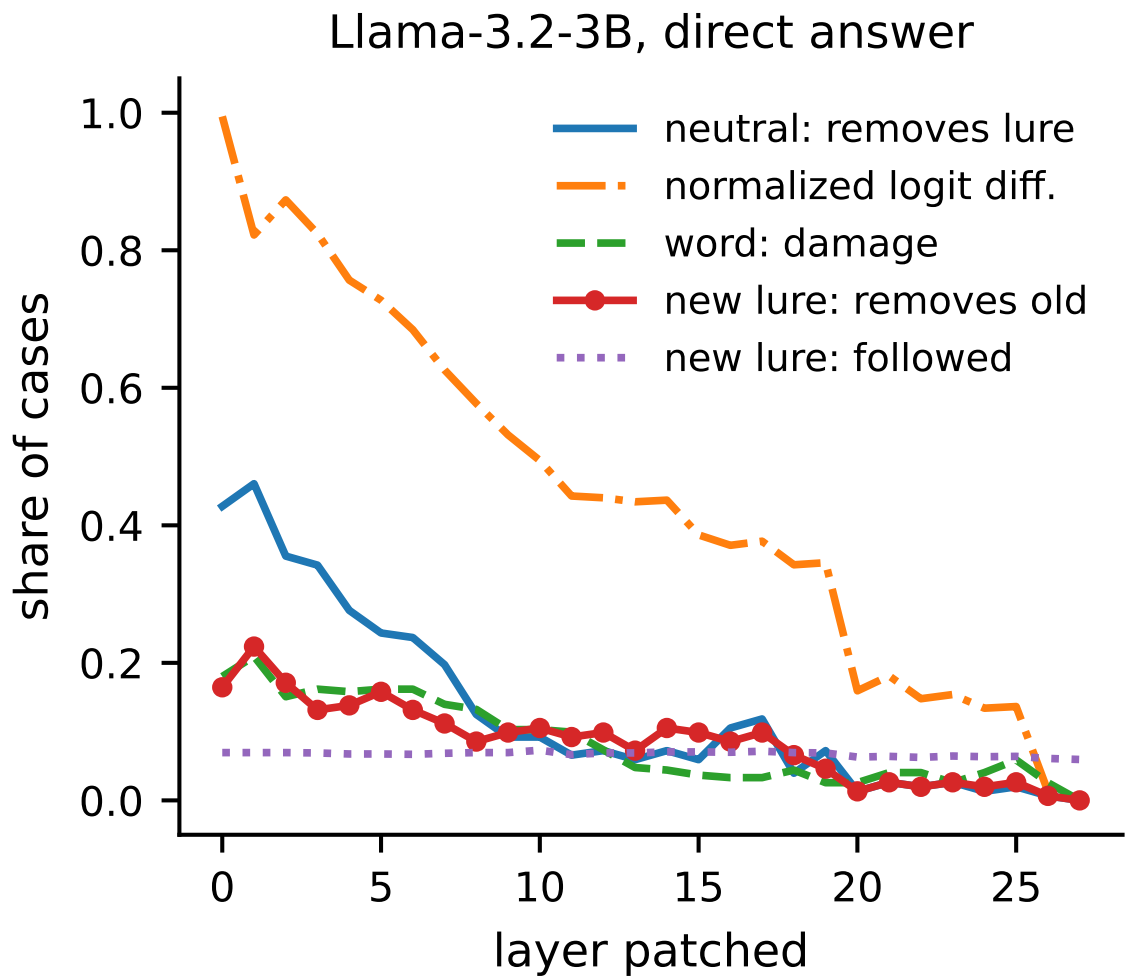


Figure 24: Lure errors removed by patching, by layer, with both controls.

Llama-3.2-3B, level 3, with chain of thought: per-token probes for the value of the queried variable (trained on neutral problems)

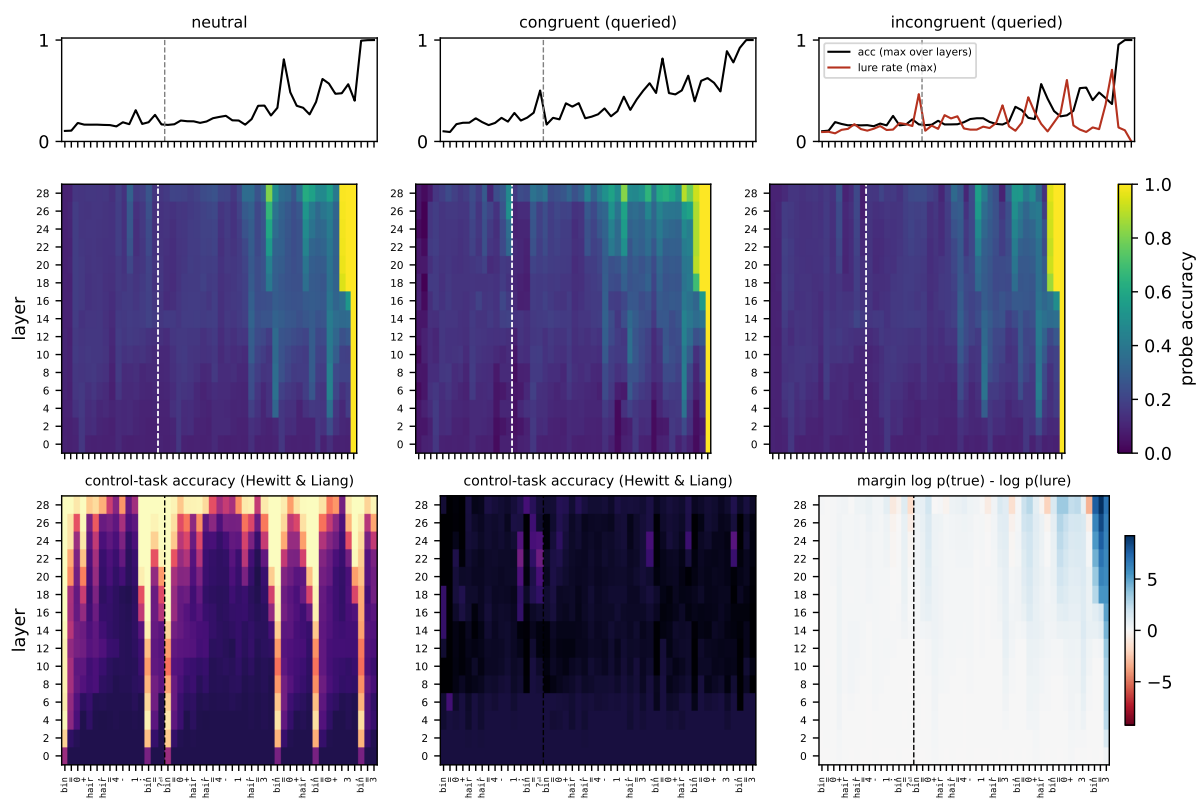


Figure 25: Accuracy heatmaps as above, chain-of-thought regime, queried variable.

Llama-3.2-3B, level 3, direct answer: per-token probes for the value of the queried variable (trained on neutral problems)

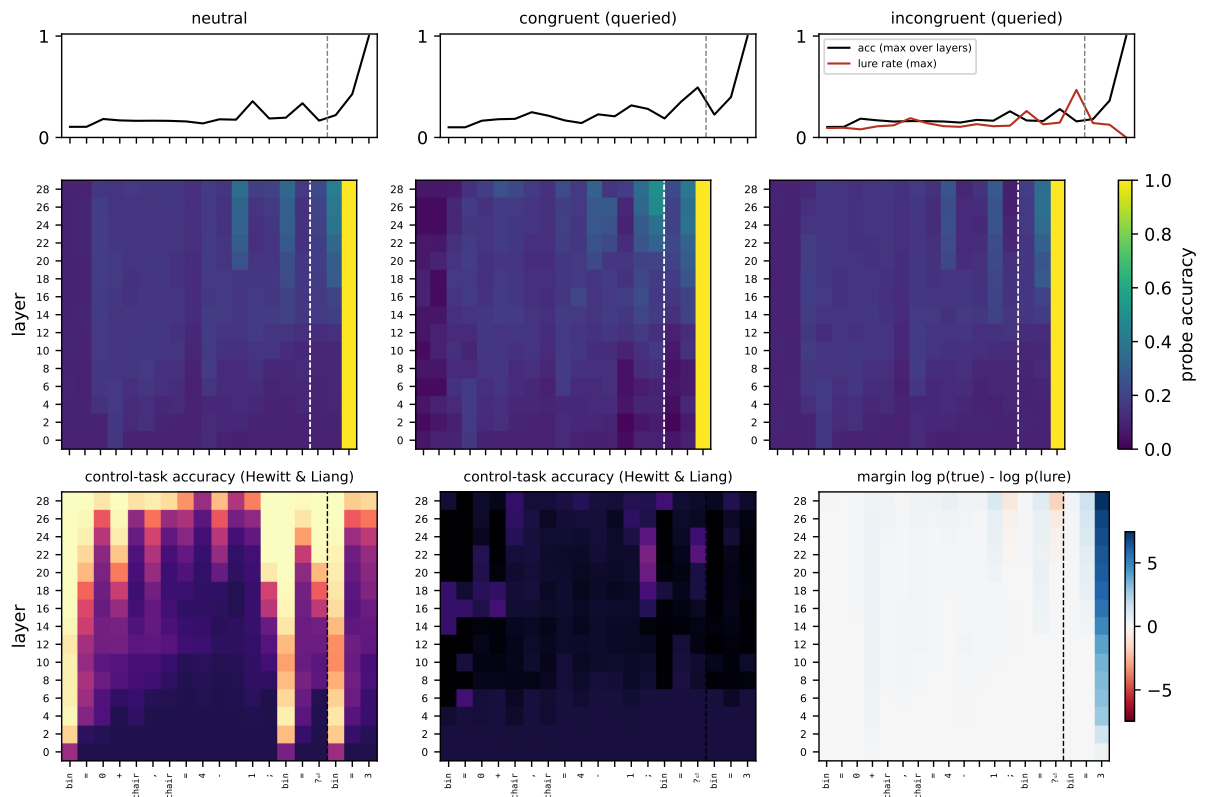


Figure 26: The same heatmaps in the direct regime, queried variable.

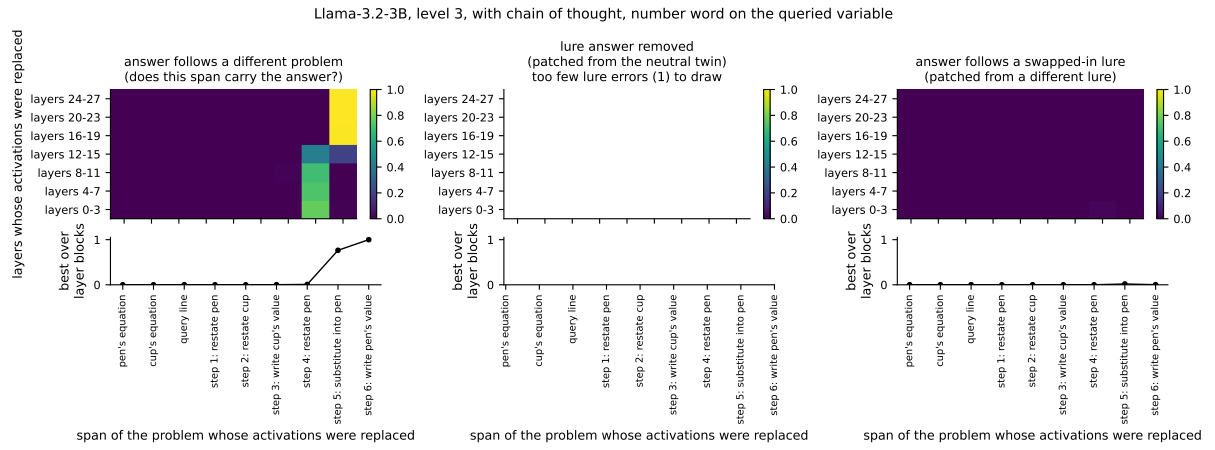


Figure 27: Span-by-layer-window patching grids, chain-of-thought regime, queried variable; the lure-removal panel is empty for the same reason.

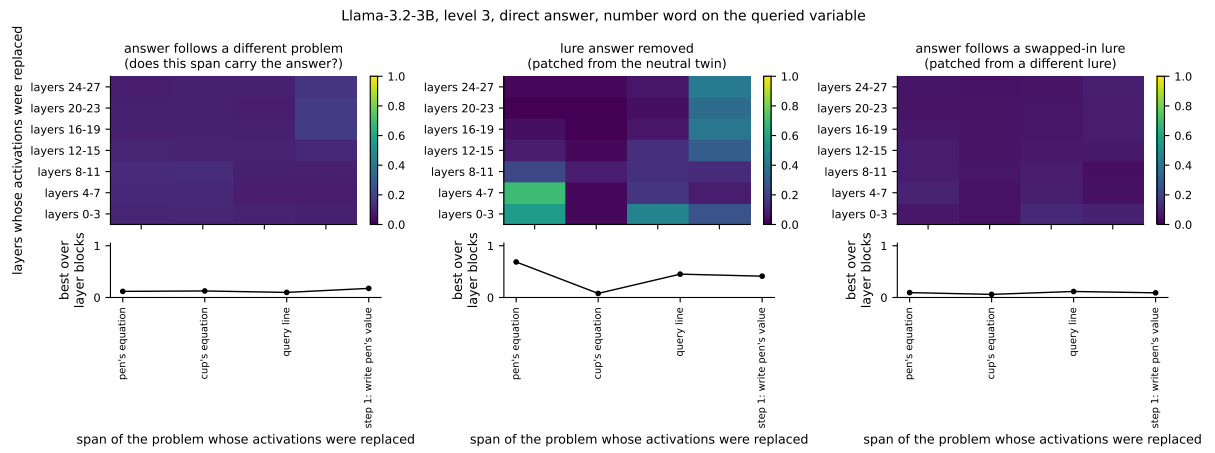


Figure 28: The same grids in the direct regime, queried variable.

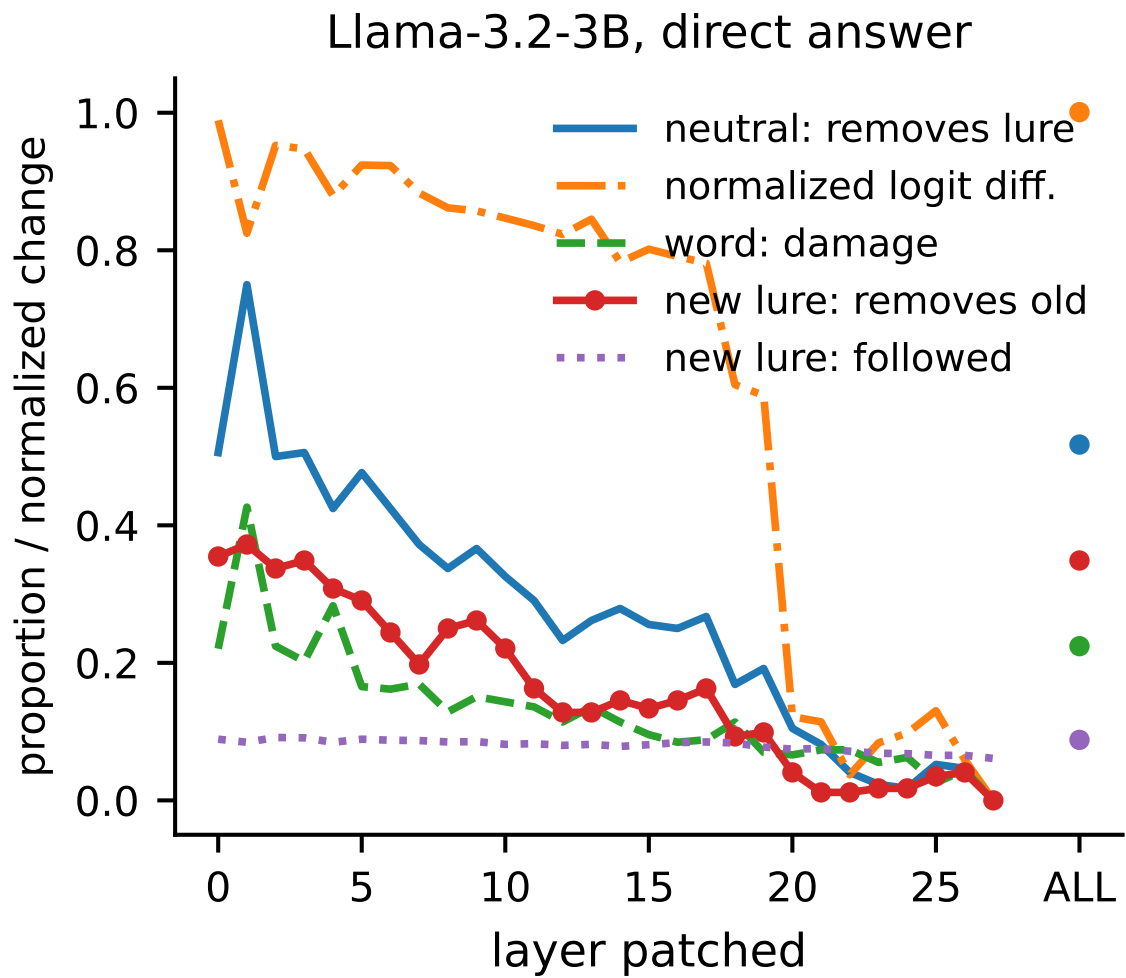


Figure 29: Lure errors removed by patching, by layer, with both controls, queried variable.

Top-1 probe prediction over the model's own chain (Llama-3.2-3B, L3, with chain of thought; probes trained at the value step of “v1” on neutral problems; 8 instances)

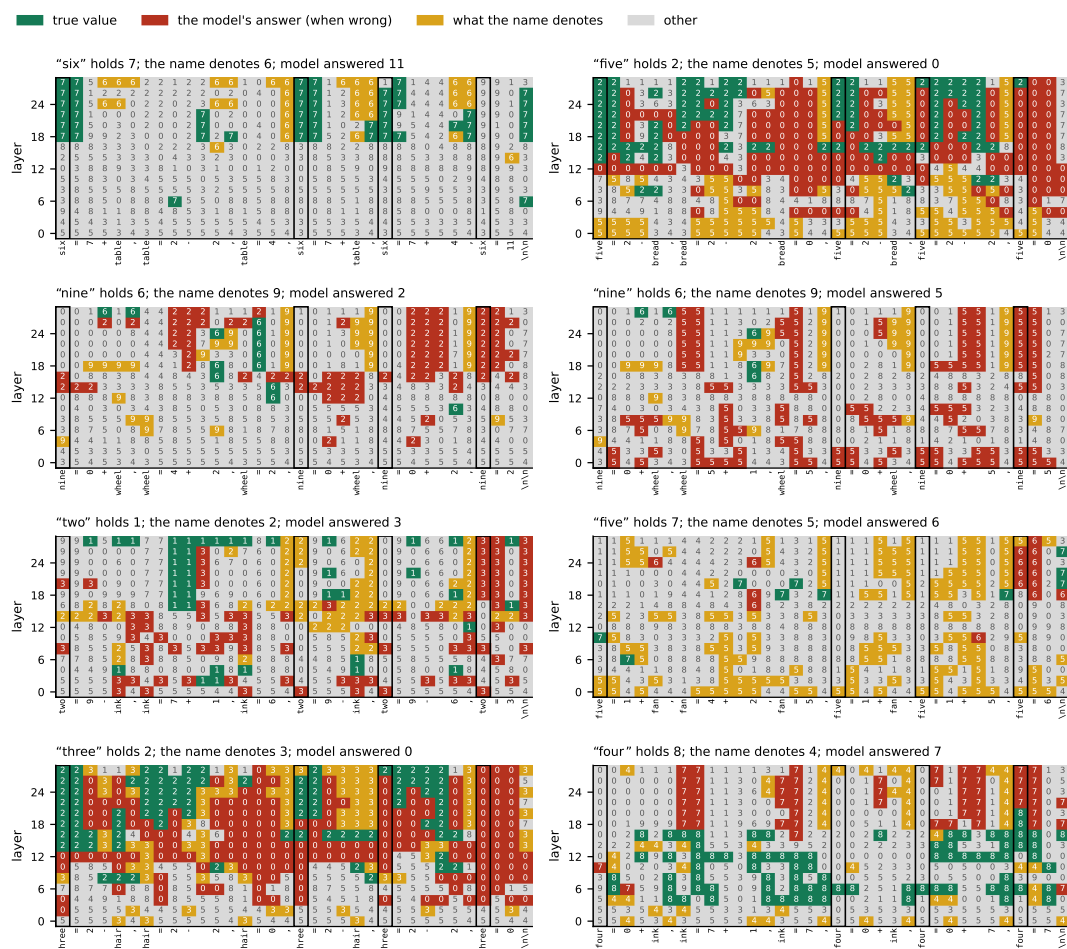


Figure 30: The own-chain probe figure at full size.

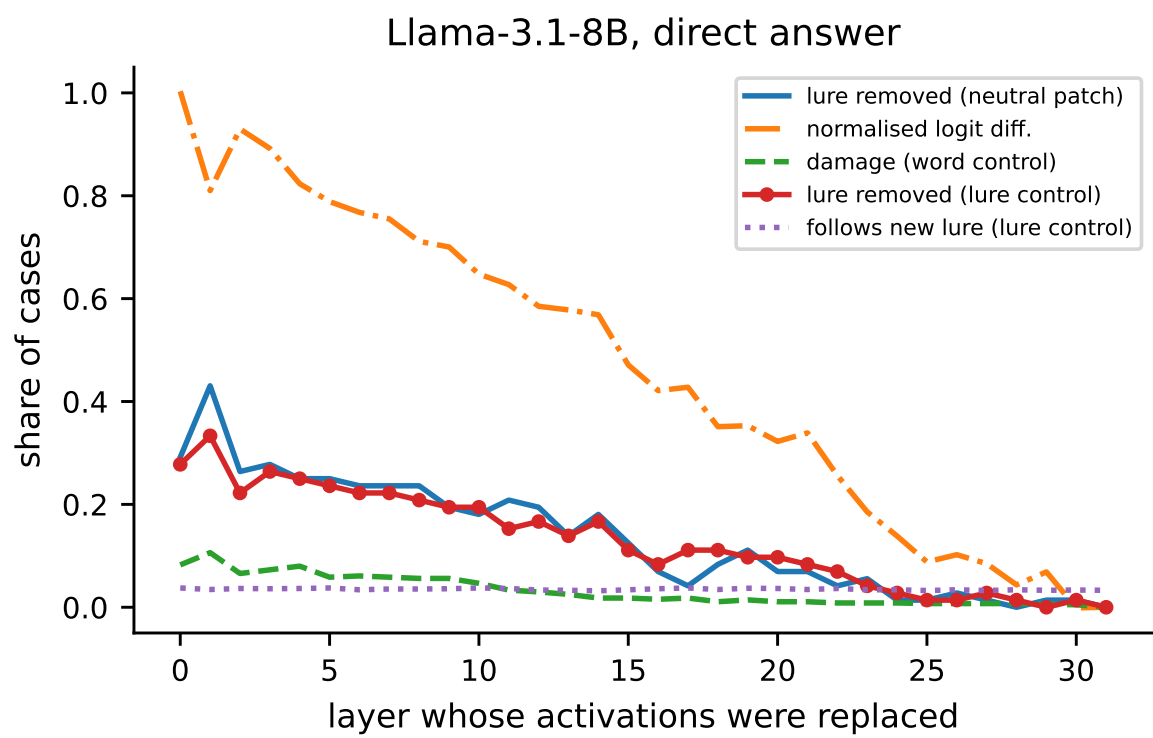


Figure 31: Llama-3.1-8B, lure errors removed by patching, by layer, with both controls, direct regime, intermediate variable.

Llama-3.1-8B, level 3, with chain of thought: per-token probes for the value of the queried variable (trained on neutral problems)

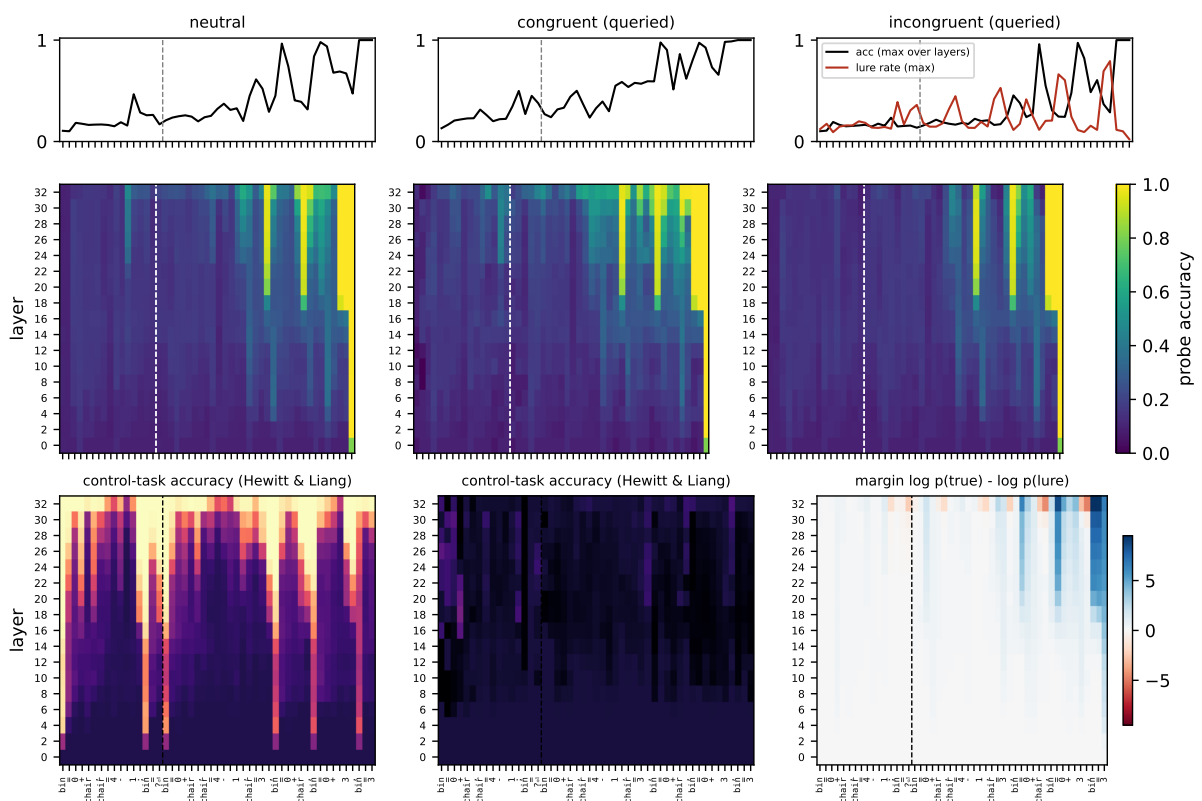


Figure 32: Llama-3.1-8B, chain-of-thought regime, queried variable: the accuracy heatmaps of the Llama-3.2-3B figures. The value is decodable from the equation that restates the variable’s definition, before the chain writes it.

Llama-3.1-8B, level 3, with chain of thought: per-token probes for the value of the intermediate variable (trained on neutral problems)

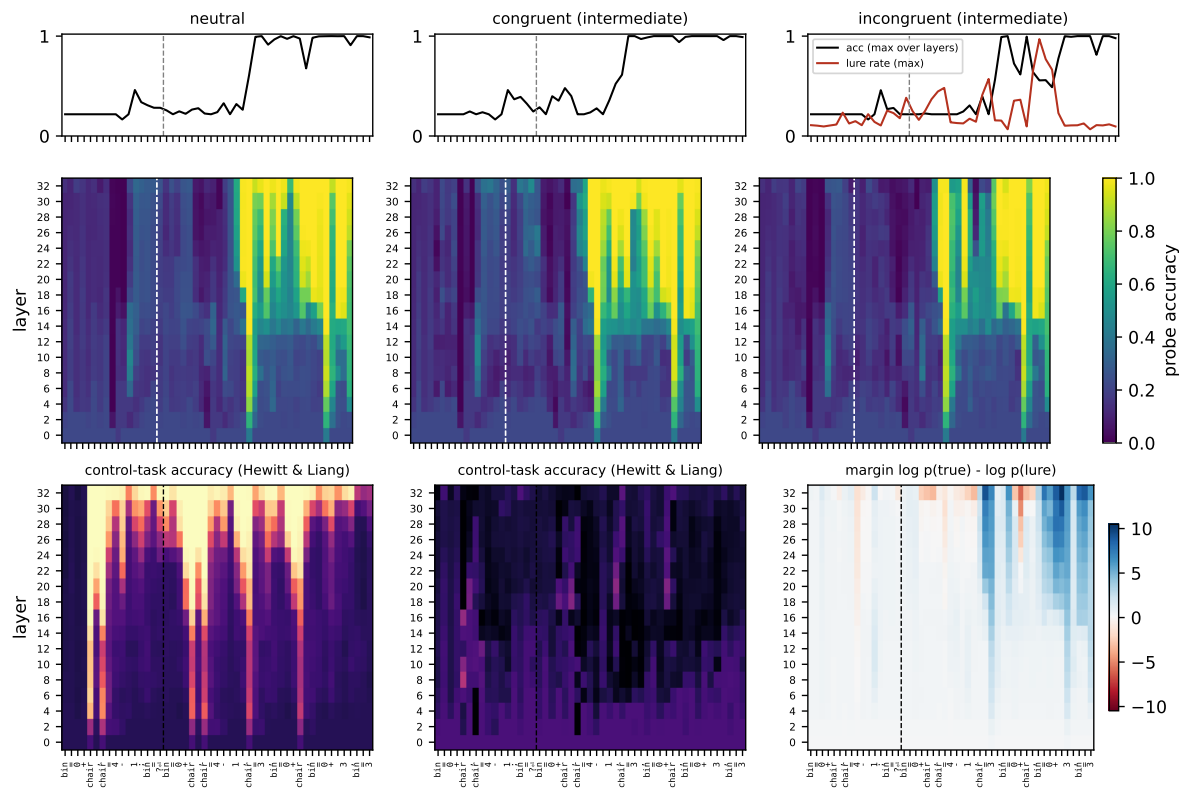


Figure 33: Llama-3.1-8B, chain-of-thought regime, intermediate variable.

Llama-3.1-8B, level 3, direct answer: per-token probes for the value of the intermediate variable (trained on neutral problems)

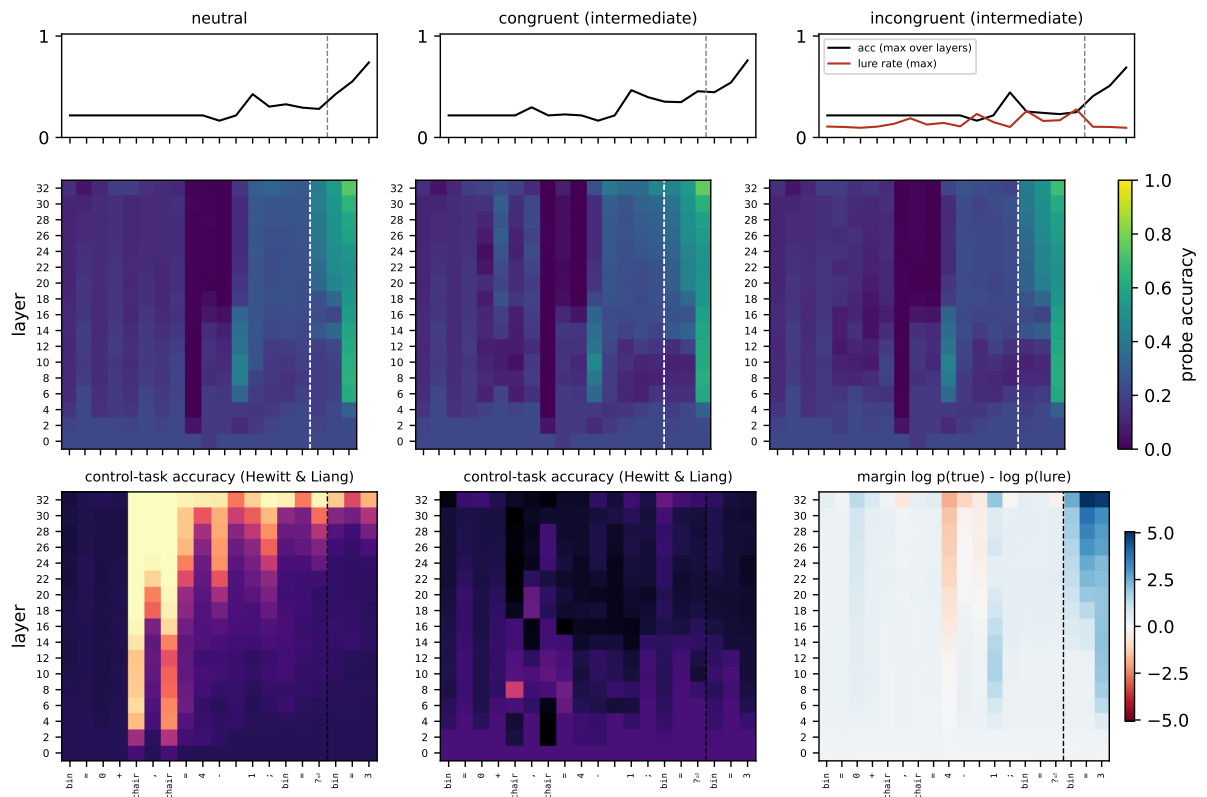


Figure 34: Llama-3.1-8B, direct regime, intermediate variable.

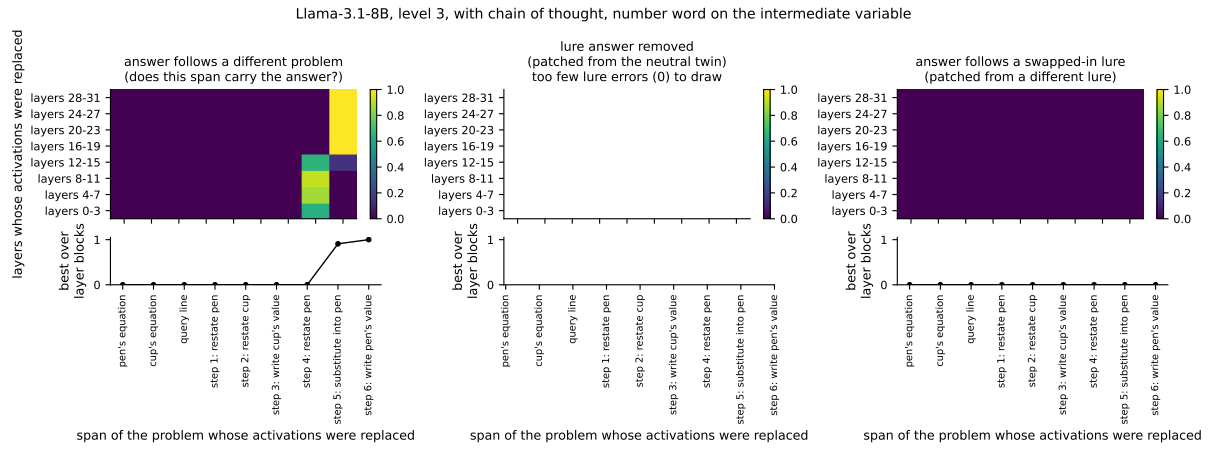


Figure 35: Llama-3.1-8B, span-by-layer-window patching grids, chain-of-thought regime. The lure-removal panel is empty: none of the 500 patched problems is answered with the lure.

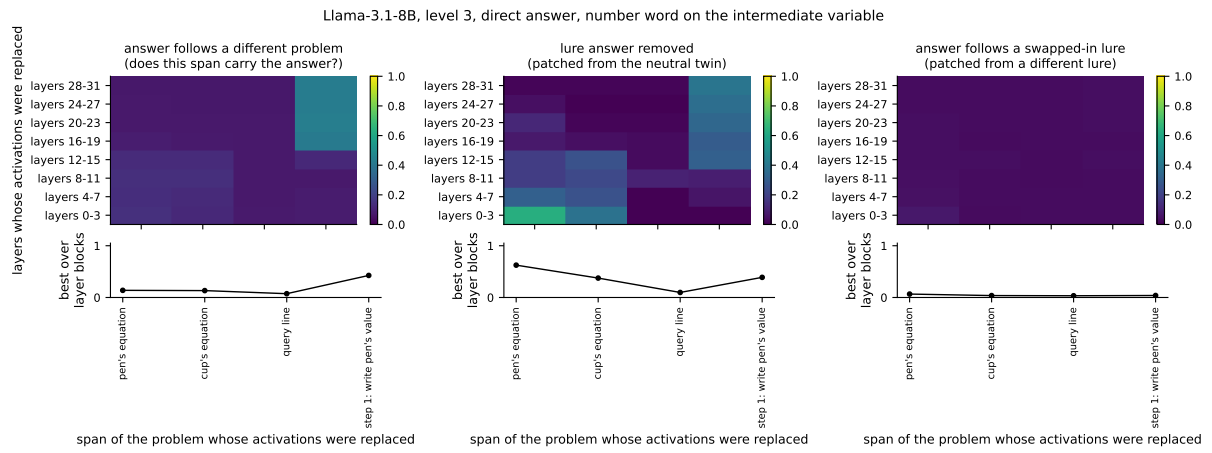


Figure 36: Llama-3.1-8B, the same grids in the direct regime.

Llama-3.1-8B, level 3, direct answer: per-token probes for the value of the queried variable (trained on neutral problems)

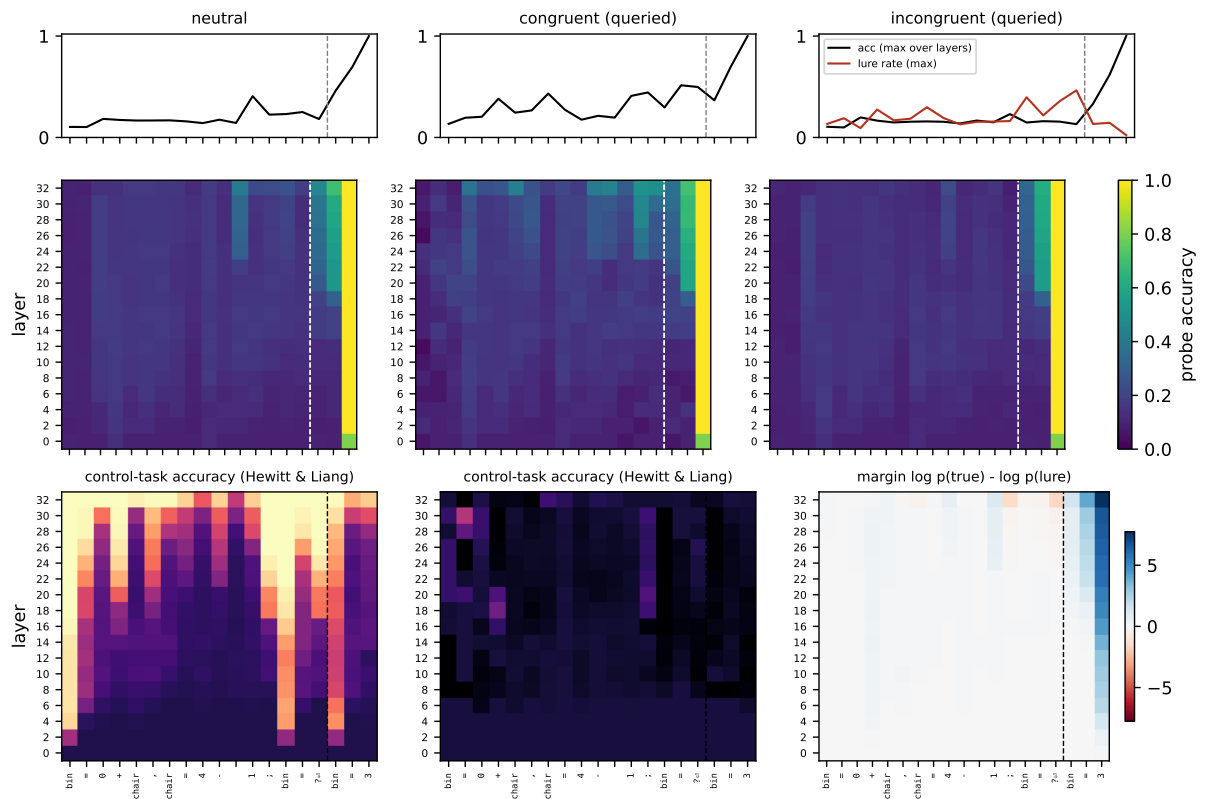


Figure 37: Llama-3.1-8B, direct regime, queried variable.

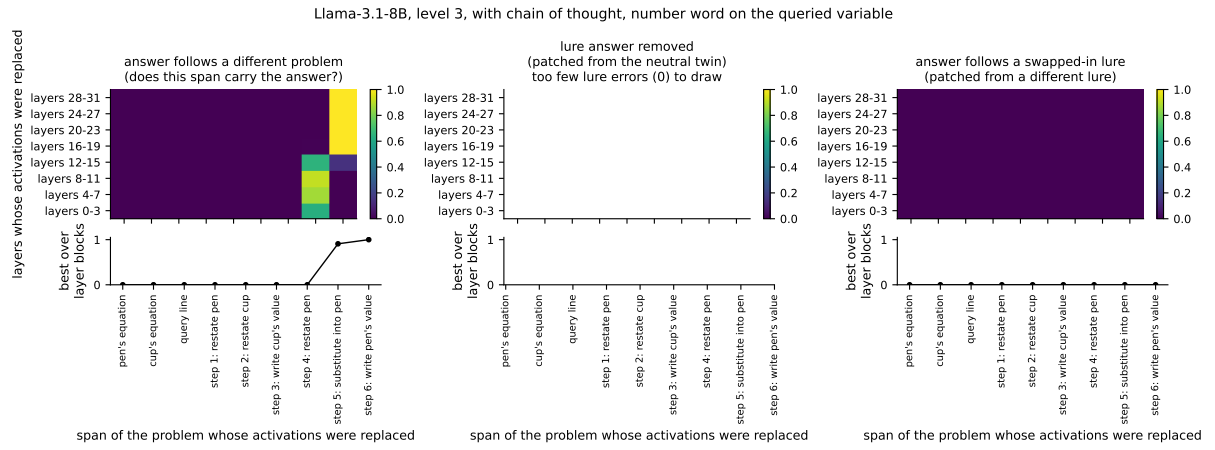


Figure 38: Llama-3.1-8B, patching grids, chain-of-thought regime, queried variable; the lure-removal panel is empty for the same reason.

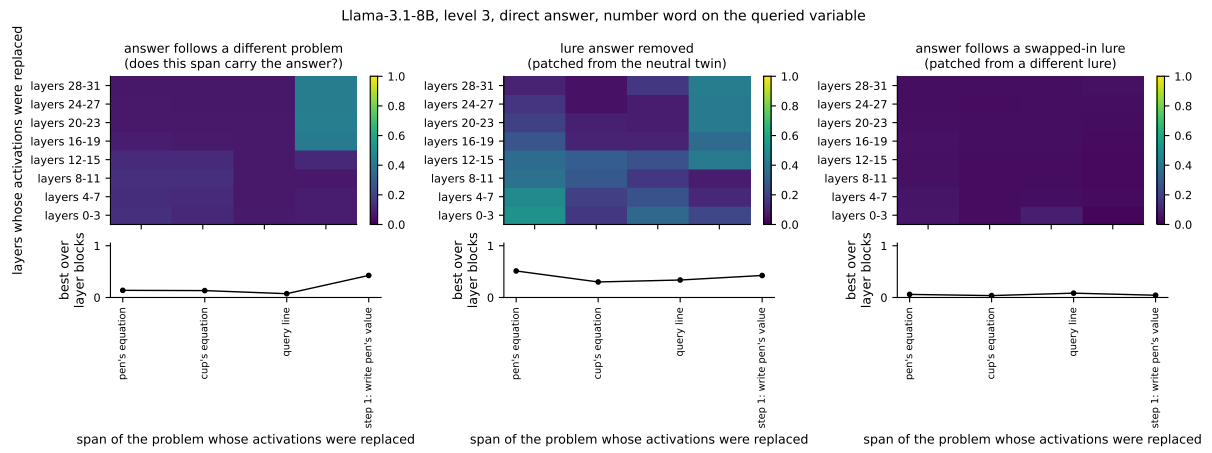


Figure 39: Llama-3.1-8B, patching grids, direct regime, queried variable.